

Provisional: descriptive bidding patterns and regulatory context

MTU15 thesis companion

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May 20, 2026

This file collects descriptive facts that any identification design must rest on: how firms bid across hour-classes (critical vs flat), how pivotal and non-pivotal firms differ, how technologies differ, and the regulatory / RT2 / reforzada context in which the reforms operate. Nothing here is an identification claim. The regression outputs from the current reform-window design are collected in a companion sibling document.

Contents

Glossary of acronyms	3
Reform-window timeline	3
Calendar coverage and same-calendar baselines	4
I Synthesis — read this first	6
1 The setting	6
2 The headline phenomenon: combined-cycle gas leaves the auction	6
3 The mechanism: stay out, get recalled, get paid more	7
4 What each reform actually did	7
5 Who responds, and how their bidding changes	7
6 What this means for identification	8
II Part A — Bidding and trading patterns	8
7 Hour-class taxonomy: critical, flat, midday	8
8 Who price-sets	13
8.1 For the day-ahead market	13
8.2 For the intraday markets	15
9 Bid-curve shape: per-curve functionals and pairwise divergences	16
9.1 Family 1 — per-curve functionals: σ_p and N_{eff}	17
9.2 Family 2 — pairwise divergences: D_w , Δp , Δq	17
9.3 Bid graphs by tech, firm, and within-hour quarter	19
9.4 Results: per-curve functionals, and the pairwise D_w diagnostic	21

9.5	Functional PCA of bid-curve shape changes across regimes	25
10	Continuous-intraday repositioning per technology	27
10.1	Seasonality-adjusted CI volumes and per-firm within-hour quarter rebalancing .	28
10.2	CI volumes relative to DA, IDA, and bilateral channels	30
10.3	CI transaction prices vs DA prices	31
III	Part B — Regulatory and confounder context	32
11	The q_1 drop: CCGT auction-cleared collapse	33
11.1	The three-component decomposition	33
11.2	Headline table and reading	35
11.3	Parallel-trends on the post-clearing gap	36
11.4	Bilateral channel as a sanity check	36
11.5	Identification implication	37
12	REE post-clearing inclusion as the absorbing mechanism	37
12.1	Three measurement paths	37
12.2	Full post-clearing intervention by <i>tipo redespacho</i> code	38
12.3	Per-firm $\times$ per-technology Path A reading	38
12.4	Per-firm CCGT post-clearing revenue (proxy)	39
12.5	CCGT marginal cost and the Fase I scarcity rent	40
13	Geographic CCGT concentration and zonal market power	41
13.1	Bid-side heterogeneity by zonal dominance	42
14	System cost cascade: the full ajuste view	43
14.1	Service inventory	44
14.2	Per-regime price comparison, by direction	44
14.3	Per-regime cost composition (REE-published cost indicators)	45
14.4	Per-day equivalents (the headline)	46
14.5	Efficiency gains across the three reform stages	47
15	Cournot reading and CNMC enforcement context	49
15.1	Cournot reading of the cap-padded ladder	49
15.2	Historical CNMC sanction precedent	49
15.3	Blackout-cohort expedientes (separate channel)	50

Glossary of acronyms

Acronym	Meaning
CCGT	Combined-cycle gas turbine: flexible, dispatchable thermal generator.
DA	Day-ahead market (Mercado Diario), cleared once per day.
IDA	Intraday auctions (3 European SIDC sessions post-2024-06-14).
MCP	Market clearing price.
MTU	Market time unit (MTU60 = hourly bidding; MTU15 = 15-minute bidding).
ISP15	Imbalance Settlement Period at 15 minutes (settlement granularity, started 2024-12-01).
REE	Red Eléctrica de España, the Spanish transmission-system operator.
OMIE	Operador del Mercado Ibérico de Energía, the wholesale market operator.
PDBF	Programa Diario Base de Funcionamiento — first post-DA viable schedule (REE-built).
PDBC	Programa Diario Base de Casación — pre-PDBF DA-cleared schedule (OMIE-built).
PHF	Programa Horario Final — post-IDA + post-RT viable schedule.
RES	Renewable Energy Sources (wind, solar PV, solar thermal, RoR hydro).
TR	Tiempo Real — real-time restrictions phase (PO 3.4–3.5), pay-as-bid.
Fase I/II	Restricciones técnicas Fase 1 (post-PDBF, pre-IDA) and Fase 2 (post-IDA), pay-as-bid.
aFRR	Automatic Frequency Restoration Reserve (Regulación secundaria), PICASSO platform.
mFRR	Manual Frequency Restoration Reserve (Regulación terciaria), MARI platform.
RR	Replacement Reserves (Reserva de Sustitución), TERRE platform.
BSP	Balance Service Provider (firm enrolled to supply balancing energy).
EUPHEMIA	Day-ahead pan-European auction algorithm (clears DA market).

Reform-window timeline

The five reform windows used throughout this document are defined as follows. The “pre-IDA window” (pre 2024-06-14) is the baseline regime, never directly compared in the main tables but referenced for context.

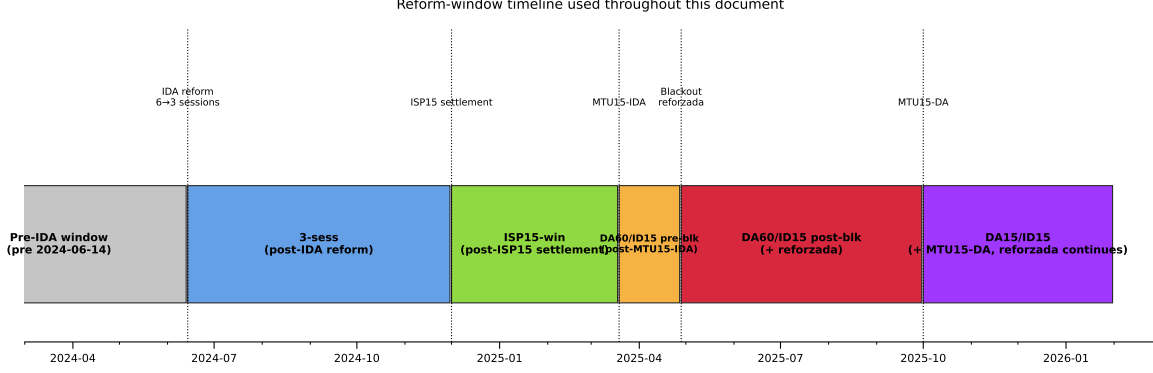


Figure 1: Reform-window timeline. Five named windows partition the post-2024-06-14 sample; the pre-IDA window is the baseline regime. Dotted vertical lines mark the five reform dates that produce the window boundaries.

Calendar coverage and same-calendar baselines

Why this matters. Spanish electricity is strongly seasonal: heating-driven demand peaks in winter, cooling in summer, hydro inflows peak in spring, solar production peaks at the summer solstice, wind blows hardest in winter. The five reform windows above do *not* span the calendar uniformly — each window covers a specific subset of calendar months, and these subsets differ across windows. Any cross-regime comparison that does not hold the calendar fixed therefore confounds the regime change with the seasonal pattern. **This applies to every variable in this document:** bid prices, D_w , Δp / Δq , redispatch volumes, imbalance MWh, system-cost levels, RES capture rates. A reader who picks any table in this document should first read Table 1 below to know which calendar months that regime covers and what comparator is available.

Table 1: Calendar coverage of each reform window, and same-calendar baseline availability. “Calendar months covered” is the set of clock months that fall inside the window. “Days” is the window length. “Same-calendar pre-IDA placebo” indicates whether a clean same-calendar baseline exists in the pre-IDA period (2018-06-14 through 2024-06-13), where no granularity reform was active. “Reforzada overlay” flags windows in which the post-blackout reinforced-operation policy (active 2025-04-28 onward) is also in effect.

Regime	Calendar dates	Months covered	Days	Reforzada?	Same-calendar pre-IDA placebo
pre-IDA	$\leq 2024-06-13$	all 12, multi-year	–	no	N/A (is the baseline)
3-sess	2024-06-14 – 2024-11-30	Jun–Nov	170	no	Clean (Jun–Nov 2018–2023)
ISP15-win	2024-12-01 – 2025-03-18	Dec–Mar	108	no	Clean (Dec–Mar 2018–2024)
MTU15-IDA pre-blk	2025-03-19 – 2025-04-27	late Mar – Apr	40	no	Clean (same 40 days 2018–2024)
MTU15-IDA post-blk	2025-04-28 – 2025-09-30	May – Sep	156	yes	Clean for granularity (May–Sep 2018–2023, pre-IDA); none for reforzada (no pre-2025-04-28 reforzada in data)
DA15/ID15	2025-10-01 – present	Oct–May (partial, ongoing)	220+	yes	None clean. 2024–25 same-calendar comparators are themselves a mix of post-MTU15-IDA + ISP15-win regimes

Three categories. Each regime falls into one of three buckets for the purpose of cross-regime

reading:

1. **Clean same-calendar placebo available.** *3-sess, ISP15-win, MTU15-IDA pre-blk.*
For each of these, we can restrict the pre-IDA baseline to the same calendar months in 2018–2024 (and average across years), and the difference (regime – pre-IDA same-cal) is interpretable as the regime effect with seasonality differenced out. Any cross-regime reading that uses these windows defaults to this same-calendar comparison unless explicitly stated otherwise.
2. **Reforzada-confounded, partial calendar-match available.** *MTU15-IDA post-blk.*
The window spans May–Sep 2025, which has clean same-calendar comparators in pre-IDA (May–Sep 2018–2023). But these comparators are pre-reforzada. So the difference (MTU15-IDA post-blk – pre-IDA same-cal) confounds (a) the granularity reform, (b) operación reforzada, and (c) the post-blackout system state. There is no calendar period in the data in which reforzada is active but MTU15 is not, so reforzada cannot be differenced out via a same-calendar pre-reform window. For descriptive purposes, May–Sep 2025 is best read as “post-MTU15-IDA + reforzada + post-blackout, jointly”; see §12 on disentangling these channels.
3. **No clean calendar match.** *DA15/ID15.* The window covers Oct 2025 onward. The same calendar months one year earlier (Oct 2024 – May 2025) are themselves a mix of pre-MTU15-IDA, ISP15-win, MTU15-IDA pre-blk, and MTU15-IDA post-blk regimes — so calendar-matching to “one year ago” does not produce a single-regime comparator. The cleanest available calendar comparators for DA15/ID15 are the same months in pre-IDA years (Oct 2023 – May 2024 was largely pre-IDA, with the IDA-3 reform starting only mid-Jun 2024; Oct 2022 – May 2023 and earlier are all pre-IDA). But even with pre-IDA same-calendar baselines, the reforzada overlay is unresolved by this comparator: DA15/ID15 is the union of MTU15-DA, MTU15-IDA, and reforzada in the same calendar window, and our data does not contain a period where any one of these three is active without the others. Therefore any DA15/ID15 reading should be flagged as descriptive only, with the three confounds explicitly named.

Fourier-deseasonalization. For a daily outcome y_t we fit a pooled Frisch-Waugh-Lovell regression on $t \in [2022-01-01, 2026-05-15]$:

$$g(y_t) = \alpha + \sum_r \beta_r \mathbf{1}\{\text{regime}(t) = r\} + \sum_{k=1}^4 [a_k c_k(t) + b_k s_k(t)] + \sum_{j=1}^6 \delta_j \mathbf{1}\{\text{dow}(t) = j\} + \varepsilon_t,$$

with omitted regime pre-IDA, link g equal to log for positive levels and logit for bounded shares, and day-of-year sinusoids $c_k(t) = \cos(2\pi k \text{doy}(t)/365)$, $s_k(t) = \sin(2\pi k \text{doy}(t)/365)$. Four harmonics (annual, semi-annual, four-month, quarterly) capture the seasonal cycle; the six day-of-week dummies absorb weekly periodicity. Seasonality is pooled across regimes because the 40-day MTU15-IDA pre-blk window cannot identify regime-specific Fourier coefficients.

Seasonality-adjusted (SA) value. $\hat{y}_r^{\text{SA}}$ is the predicted level of y in regime r at annual-mean Fourier (zero by orthogonality) and within-week DOW mean, with a Duan-smearing correction across in-sample residuals $\{\hat{e}_i\}$ for the non-linear link:

$$\hat{y}_r^{\text{SA}} = \frac{1}{n} \sum_{i=1}^n g^{-1} \left(\hat{\alpha} + \hat{\beta}_r + \frac{1}{7} \sum_{j=1}^6 \hat{\delta}_j + \hat{e}_i \right).$$

In tables, SA values sit in [coloured brackets](#) next to the raw regime mean (omitted where the value is calendar-matched by construction). Where raw and SA move the same way the pattern

survives seasonality control; where they diverge, only the SA value is comparable across regimes. For the reforzada-confounded windows (MTU15-IDA post-blk, DA15/ID15) the SA value still bundles granularity, reforzada, and the post-blackout system state — it differences out the calendar, not the policy overlap.

Reading guide. Part A is about bidding and trading patterns. Part B is about the regulatory/post-clearing channels (reforzada, geographic dominance, CNMC sanctions, etc.) that are independent of the granularity reform but constrain how the identification can be set up.

Part I

Synthesis — read this first

Executive summary: the phenomenon, the mechanism, and what each reform did, written for a reader who has not yet seen Parts A and B. Every claim points to the figure or table in those parts that sustains it. Numbers tagged (SA) are seasonally adjusted; (cal-matched) are same-calendar-month comparisons; untagged numbers are raw.

1 The setting

- **Four market reforms in sixteen months:** intraday auctions $6 \rightarrow 3$ European sessions (Jun 2024); 15-min imbalance settlement (ISP15, Dec 2024); 15-min intraday auctions (MTU15-IDA, Mar 2025); 15-min day-ahead (MTU15-DA, Oct 2025). Calendar map: Table 1.
- **Cutting across them:** the 28 April 2025 Iberian blackout and the *operación reforzada* stance that followed — the system operator keeps voltage-fragile-zone gas plants physically running whether or not they cleared the market.
- **The intuition for the whole document:** the reforms are a *granularity* story, reforzada is a *market-power* story; they overlap in calendar time, and everything here is about telling the two apart.

2 The headline phenomenon: combined-cycle gas leaves the auction

- **The four big combined-cycle gas (CCGT) operators stopped clearing the day-ahead market.** DA-cleared volume fell 83% same-calendar-month, Dec 2024 $\rightarrow$ Dec 2025 (1,444 $\rightarrow$ 244 GWh, Table 23); every other technology held flat or grew.
- **The intuition: the gas did not stop generating — it stopped generating *through the auction*.** The output reappears on a different channel.
- The “post-DA gap” (final program minus day-ahead-plus-bilateral) rose $\sim +1,044$ GWh/month for CCGT after the blackout, and *fell* for every other technology on the same channel (Table 25).
- Deseasonalised, the CCGT post-DA gap goes ~ 47 GWh/day (3-sess baseline) $\rightarrow \sim 389$ GWh/day (post-blackout) (SA) — a structural break, not a seasonal wobble. The gas plants increasingly get dispatched by the operator *after* the auction clears, rather than competing inside it.

3 The mechanism: stay out, get recalled, get paid more

Why would a profit-maximising plant walk away from the market it exists to sell into? Because being recalled afterwards pays better. Four observable steps:

- **Stay out.** Naturgy’s CCGT fleet bids $\sim 840/\text{MWh}$ in every zone — dominant or not, 2024 and 2025 (cal-matched, Table 31). Far above any plausible clearing price: a bid engineered *not* to clear.
- **Get recalled.** The operator adds the plant back via post-clearing redispatch under reinforced-operation security criteria. That leg — “Fase I” — is paid *pay-as-bid*, not at a uniform market price (§12): a locally-indispensable plant names its own price on the recall.
- **Collect the markup.** Fase I-up averaged 168/MWh under DA15/ID15 vs a bottom-up CCGT marginal cost of 100–110 — a 30–60% markup over cost. And it *widened* across the reform sequence: 44–59/MWh premium at 3-sess $\rightarrow$ 58–68 at DA15/ID15 (Table 29).
- **Concentrated, and not new.** Naturgy alone captures $\sim$ half of Big-4 post-clearing CCGT revenue (Table 28); per-day take $\sim 2\text{M}$ pre-reforzada $\rightarrow \sim 5\text{--}8\text{M}$ post. The CNMC sanctioned Naturgy for exactly this practice, same southern-corridor plants, in 2016–2017 (Table 36, expediente SNC-DE-175-17). *The reforms did not invent the strategy — they changed which channel makes it pay.*

4 What each reform actually did

- **ISP15 (settlement, Dec 2024) — the cleanest to read**, nothing else changed in its window. 15-min imbalance settlement mechanically shrinks the within-hour variance that gets settled: imbalance volume per cell fell $\sim 60\%$ — ~ 50 pp mechanical variance-averaging + ~ 10 pp genuine behavioural improvement in balance-responsible-party forecasting (§14.5).
- **MTU15-IDA (intraday, Mar 2025) — the biggest genuine efficiency gain.** 15-min intraday products mean within-hour scarcity gets resolved *in the auction* instead of bleeding into real-time emergency redispatch. Real-time technical-restriction cost: $\sim 5.8\text{M}/\text{day} \rightarrow \sim 2.9\text{M}/\text{day} \rightarrow \sim 1.2\text{M}/\text{day}$ (SA) by DA15/ID15 (Table 35, Fig. 17).
- **MTU15-DA (day-ahead, Oct 2025) — the hardest to read**, its window sits entirely inside reforzada. Small extra efficiency ($\sim 1.5\text{M}/\text{day}$ real-time + reserve savings) but the window also carries reforzada-driven Fase I cost growth ($+2.5\text{M}/\text{day}$). Net per-day adjustment cost *rises* — not because granularity failed, but because the reinforced-operation stance is expensive and shares the calendar window. Fig. 17 shows it: MTU15 channels shrink bar by bar while the Fase I channel grows. The two are logically distinct; we never report their sum as “the effect of MTU15-DA”.

5 Who responds, and how their bidding changes

- **The strategic actor is combined-cycle gas**; granularity handed it a new margin. Before MTU15, one price-quantity schedule per clock-hour; after, four within-hour schedules it can shape differently.
- We measure that shaping two ways (§9.4): *per-curve* metrics — the price spread σ_p and effective tranche count N_{eff} of each single bid curve — and the *pairwise* D_w dissimilarity between a unit’s four within-hour quarter curves near MCP. For CCGT in critical hours,

the share of cells with non-trivial within-hour shaping ($D_w > 0$) roughly *doubles* across the three MTU15-IDA regimes (Table 10); hydro and pumped hydro move the same direction, weaker. *The granular market created a new pricing margin, and the technologies with market power in the high-stakes hours exercised it.*

- **No longer a one-firm story:** under DA15/ID15 all four big CCGT firms shape their within-hour bids (HC 79%, GE 75%, GN 72%, IB 59% of critical-hour cells); before the blackout it was concentrated at Naturgy.
- **Caveat for the regression design:** the critical-versus-flat asymmetry — the backbone of the identification — still holds in direction (CCGT 71% critical vs 51% flat under DA15/ID15) but is now a matter of *degree*, not a clean treated-versus-untreated split: flat hours are no longer “unshaped”.

6 What this means for identification

- The three granularity reforms overlap in time with reforzada; the MTU15-DA window sits *entirely* inside it. No single regression can be read as “the effect of MTU15-DA” without conflating a genuine granularity efficiency with a reforzada-driven cost.
- **Strategy** (companion document `regression_results.tex`): choose reform windows in which reforzada is held constant, and read each outcome one at a time rather than as one bundled effect.
- **External check the mechanism is real** (not a measurement artefact): the CNMC’s 2019 sanction (SNC-DE-175-17) and its 2026 expediente cohort pursue precisely the stay-out-and-get-recalled conduct documented here.

The rest of the document is the evidence base. Part A documents the bidding and trading patterns one technology, one firm, one hour-class at a time; Part B characterises the regulatory and post-clearing channels (reforzada, geographic dominance, CNMC enforcement) that surround the granularity reforms and constrain how the identification can be set up.

Part II

Part A — Bidding and trading patterns

Reading guide for Part A. Three setup sections come first. §7 defines the **hour-class taxonomy** (critical / flat / midday) used to slice every table and figure that follows. §8 characterises **who price-sets** in the day-ahead market (§8.1) and, methodologically, in the intraday markets (§8.2). §9 then defines the **bid-shape metrics** in two families: *per-curve functionals* (one scalar per single bid curve — σ_p , N_{eff} , the fPCA scores) and *pairwise divergences* (a statistic of two curves — D_w , Δp , Δq). Visual results (per tech $\times$ firm $\times$ quarter bid curves with mean clearing prices overlaid) come in §9.3; quantitative tables come in §9.4.

7 Hour-class taxonomy: critical, flat, midday

Empirical classification: within-hour standard deviation of residual demand. The crit/flat/midday split is **not assumed** on demand-level grounds (“critical hours are the ones with the highest demand”). It is derived from the within-hour *variability* of residual demand. The intuition is mechanical: *MTU15 granularity has economic content only where the within-hour profile actually moves*. If residual demand is essentially flat within a clock-hour, the hourly

clearing price is already a good summary; granular per-quarter pricing has no signal to add. If residual demand changes sharply within the hour (morning ramp, evening peak, solar fade), the hourly clearing price was a compromise across rapidly-changing within-hour conditions; granular pricing unlocks the ability to discriminate across quarters and is therefore where strategic bidding can have a quarter-specific footprint.

Concretely, for clock-hour h on date d , let $D_{h,q,d}$ be the residual demand ($= \text{load} - \text{wind} - \text{solar}$) at quarter $q \in \{1, 2, 3, 4\}$ of that hour. The hour’s quarter-mean is $\bar{D}_{h,d} = \frac{1}{4} \sum_q D_{h,q,d}$. We compute, per (date, hour), the within-hour standard deviation

$$\sigma_{\text{within}}(h, d) = \sqrt{\frac{1}{4} \sum_{q=1}^4 (D_{h,q,d} - \bar{D}_{h,d})^2},$$

and take the expectation over all 2025 dates to get a clock-hour metric

$$\sigma_{\text{within}}(h) = \mathbb{E}_d[\sigma_{\text{within}}(h, d)] \quad [\text{MW}].$$

The 24-hour ranking of $\sigma_{\text{within}}(h)$ is the actual driver of the classification. Hours with high σ_{within} are critical; hours with low σ_{within} are flat; midday hours are excluded for a separate reason (below). In economic terms, high σ_{within} means the within-hour residual-demand curve moves enough that equilibrium quarter prices and quantities should differ; hour-level pricing was a binding pooling constraint, and granular pricing relaxes it. Strategic firms with within-hour market power therefore have scope to exploit it in critical hours and essentially no scope in flat hours — which is the heterogeneity Part A’s CCGT bid-shape analysis is set up to detect.

Data, window, and frequency. $D_{h,q,d}$ is computed from two ENTSO-E Transparency Platform sources, both at 15-minute ISP resolution:

- Spanish system load: ENTSO-E A65 (ENTSO-E A65 load series), filtered to MTU = 15 minutes.
- Spanish wind + solar generation: ENTSO-E A75 (ENTSO-E A75 wind+solar actual generation series), PSR types B16 (solar) + B18 (wind offshore) + B19 (wind onshore), filtered to MTU = 15 minutes.

Residual demand is the per-ISP arithmetic difference $D = \text{load} - \text{solar} - \text{wind}$.

Window. The classifier runs on 2023-01-01 through 2026-05-14 (≈ 3.4 years), the largest window for which both A65 and A75 are uniformly available at 15-min granularity (pre-2023 the series are 60-min, so within-hour variance is undefined). The classifier is a structural feature of the within-day pattern, not a regime-specific statistic; all dates are pooled (weekday + weekend) and the weekday-only partition produces the same class assignments.

The 24-hour ranking. Table 2 is reproduced from the main paper’s Appendix A.7. It shows the per-hour mean load, solar, wind, residual demand, σ_{within} , the coefficient of variation $\text{CV} = \sigma_{\text{within}} / \text{residual demand}$, and the signed within-hour load change $\Delta \text{load} = \text{load}_{q_4} - \text{load}_{q_1}$, averaged over all dates 2023-01-01 through 2026-05-14. The critical-flat-midday classification falls out of σ_{within} directly.

Why the CV column. Both the *absolute* within-hour swing (σ_{within}) and the *level* of residual demand it sits on matter, and they need not rank hours the same way. The CV — the swing normalised by the level — makes the difference visible. Two readings stand out. First, the evening-peak hours 19h–22h are critical on σ_{within} (522–781 MW) but have a *low* CV (2.3–3.9%), because residual demand is at its daily maximum (~ 20 –23 GW) and divides the swing down — their within-hour curve moves a lot in MW but little relative to its own height. Second, the morning-ramp and evening-transition hours (08h, 16h) carry the highest CV (7.5–9.0%): a

large swing on a comparatively thin residual-demand base, the configuration in which a within-hour price-setter has the most relative room to discriminate. The classifier remains σ_{within} — the CV is shown as complementary context, not as an alternative threshold — but the CV column flags that “critical” is not a uniform category: the morning ramp and the evening peak are critical for different reasons.

Table 2: Hour-of-day classification metrics, Spain 2023–2026 (ENTSO-E A65 + A75, 15-min ISP, $n \approx 3.4$ years of daily observations per hour). The classifier is σ_{within} (column 6, MW). Critical hours: $\sigma_{\text{within}} \in [522, 1303]$. Flat hours: $\sigma_{\text{within}} \in [336, 357]$. Midday: $\sigma_{\text{within}} \in [468, 502]$ — intermediate, but excluded for the separate reason discussed below. Dropped (transition) hours sit at the regime boundaries. *Year-stability:* across the 3 full years 2023, 2024, 2025 (computed separately), the critical-set min σ_{within} across years is 530 MW, exceeding the flat-set max (387 MW) by 1.4 $\times$; no within-class swaps across years.

Clock-hour	Demand (GW)	Solar (GW)	Wind (GW)	Residual demand (GW)	σ_{within} (MW)	CV (%)	Δ demand (MW)	Class
00:00–01:00	22.8	0.1	7.3	15.4	412	2.7	−834	<i>dropped</i>
01:00–02:00	22.0	0.1	7.1	14.8	347	2.4	−470	<i>flat</i>
02:00–03:00	21.6	0.1	6.9	14.6	336	2.3	−190	<i>flat</i>
03:00–04:00	21.5	0.1	6.7	14.7	357	2.4	176	<i>flat</i>
04:00–05:00	22.4	0.1	6.7	15.7	587	3.8	1114	<i>dropped</i>
05:00–06:00	24.4	0.2	6.7	17.5	721	4.1	1743	<i>critical</i>
06:00–07:00	26.5	1.5	6.6	18.4	915	5.0	1423	<i>critical</i>
07:00–08:00	27.7	4.9	6.2	16.6	1078	6.5	627	<i>critical</i>
08:00–09:00	28.3	9.1	5.8	13.4	1006	7.5	253	<i>critical</i>
09:00–10:00	28.3	12.0	5.5	10.9	753	6.9	−104	<i>dropped</i>
10:00–11:00	28.2	13.3	5.4	9.5	499	5.3	−66	<i>dropped</i>
11:00–12:00	28.2	13.8	5.5	8.9	468	5.3	94	<i>dropped</i>
12:00–13:00	28.3	13.8	5.7	8.8	502	5.7	−104	<i>dropped</i>
13:00–14:00	27.9	13.5	5.8	8.5	485	5.7	−331	<i>dropped</i>
14:00–15:00	27.5	12.9	6.1	8.5	469	5.5	−122	<i>dropped</i>
15:00–16:00	27.6	11.7	6.4	9.6	677	7.1	291	<i>dropped</i>
16:00–17:00	28.2	9.0	6.8	12.3	1109	9.0	553	<i>critical</i>
17:00–18:00	29.2	5.4	7.4	16.4	1303	7.9	891	<i>critical</i>
18:00–19:00	30.5	2.2	7.7	20.5	1065	5.2	884	<i>critical</i>
19:00–20:00	31.3	0.6	7.9	22.8	522	2.3	176	<i>critical</i>
20:00–21:00	30.4	0.3	7.9	22.2	622	2.8	−1444	<i>critical</i>
21:00–22:00	28.0	0.2	7.9	20.0	781	3.9	−1851	<i>critical</i>
22:00–23:00	25.8	0.2	7.7	18.0	664	3.7	−1407	<i>critical</i>
23:00–00:00	24.2	0.1	7.5	16.5	510	3.1	−1062	<i>dropped</i>

Classification rule. The partition is read off the table by contiguous-block structure: **critical** = the morning-ramp block (5h–8h) + the evening-peak block (16h–22h), 11 hours all with $\sigma_{\text{within}} \in [522, 1303]$ MW; **flat** = the overnight block (1h–3h), 3 hours all with $\sigma_{\text{within}} \in [336, 357]$ MW; **midday** = the solar-peak block (11h–14h), 4 hours with $\sigma_{\text{within}} \in [468, 502]$ MW and residual demand < 9 GW. The 6 remaining hours (0, 4, 9, 10, 15, 23, $\sigma_{\text{within}} \in [412, 753]$ MW) straddle the thresholds and are dropped. No clustering or supervised threshold tuning is involved — the partition can be reconstructed from the sorted table by visual inspection.

Why we distinguish critical from flat. Critical-hour σ_{within} is roughly 1.5–4 $\times$ that of flat hours (522–1,303 MW vs 336–357 MW), reflecting fast within-hour shifts in residual demand: morning demand ramp + solar rise at 5h–9h, solar fade + evening peak at 16h–22h. Granular per-quarter pricing in critical hours can in principle discriminate prices across quarters that face very different supply/demand conditions, so it is where strategic bidding has scope. Flat hours have nearly-flat residual demand within the clock-hour and granular pricing has no economic content to extract — the four quarter prices, absent friction, should be essentially the same. This is why flat is the within-day control: the (crit − flat) differential strips out any day-level

shock (gas, weather, regulation) that affects both hour-classes proportionally, while flat itself carries no granularity-exploitation signal.

Falsifier already documented in the main paper. Main paper §2.4: CCGT price-setting share is 9.1% critical vs 9.0% flat (almost identical), so TTF/gas-cost shocks cannot propagate asymmetrically across the two hour-classes. The within-day differencing therefore identifies behaviour, not cost shocks.

Outcome-side validation: post-MTU15-DA actually price-discriminates more in critical hours. The classifier is a structural proxy: high σ_{within} should predict where granular pricing delivers measurable price dispersion. Post-MTU15-DA (2025-10-01 to 2026-05-14, the only window with 15-min DA quarter prices), the within-hour DA quarter clearing-price standard deviation averages **6.58 EUR/MWh in critical hours vs 2.13 in flat hours ($3.1\times$ ratio)** — midday is even tighter (1.51 EUR/MWh, solar-marginal regime). The classifier correctly identifies the hours where post-reform quarter prices actually diverge.

Why we exclude midday from the control. Midday hours (11–14) have $\sigma_{\text{within}} \in [468, 502]$ MW, intermediate between critical and flat. By the variability criterion alone, midday is neither clearly “treated” (granularity has some scope) nor clearly “control” (granularity has zero scope). What pushes midday into a third bucket is the *level* of residual demand and the implied marginal-pricing technology: midday solar peak drives residual demand down to 8.5–8.9 GW (vs ~ 15 –23 GW outside midday), and the marginal price-setting technology shifts from CCGT/hydro to solar PV + storage. Pooling midday into flat as a control would absorb both (a) the strategic-bidding signal we want and (b) the operational solar-vs-thermal differential (a separate channel that has its own dynamics). We exclude midday from the control set entirely. It appears in some descriptive overlays (channel-level histograms of Δp , Δq show crit/midday/flat side by side for comparison), but it is never used as control in the DDD.

Why a 6-hour transition gap. Hours 0, 4, 9, 10, 15, 23 sit at the boundaries between regimes. Their σ_{within} values are 412, 587, 753, 499, 677, 510 MW — straddling the flat/critical thresholds. Rather than force a binary classification, we drop them. The 14-hour analytic sample (11 crit + 3 flat) is the analytic universe for every table in Part A; the 4 midday hours are reserved for descriptive overlay only.

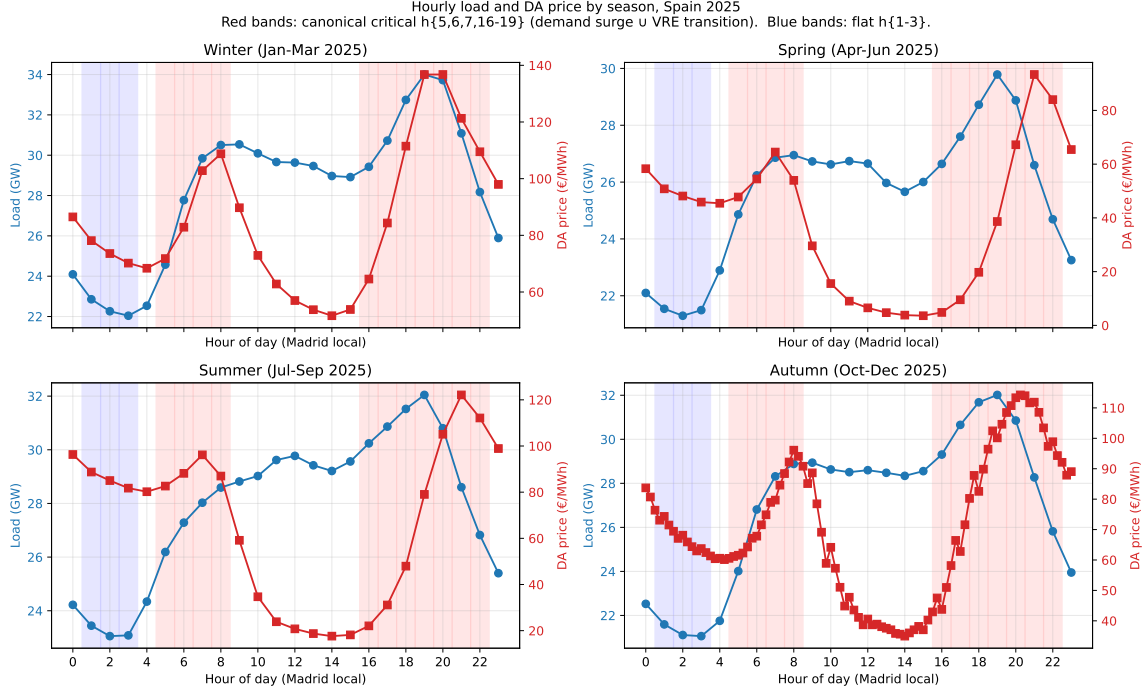


Figure 2: Hourly load and DA price by season, Spain 2025. Blue: demand; red: DA clearing price. Red bands: critical hours (05:00–09:00, 16:00–23:00); blue bands: flat hours (01:00–04:00). *Note:* demand level is shown here for descriptive context only — the formal classifier is σ_{within} of residual demand (see Table 2), not demand level. The seasonal-stability of the within-day pattern is the relevant point. Reproduced from main paper §2.1.

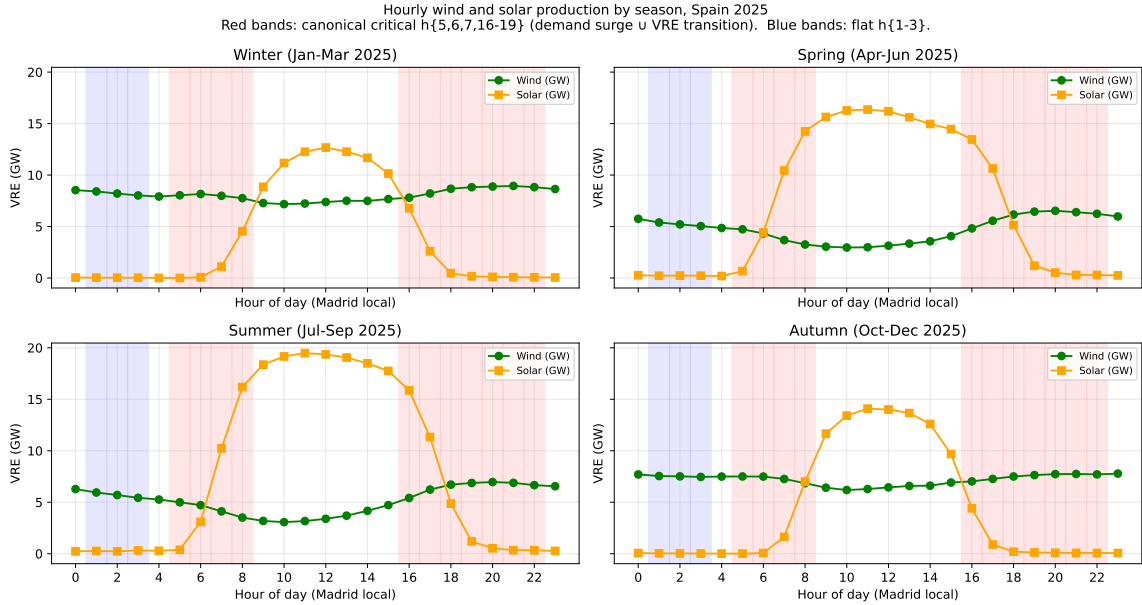


Figure 3: Hourly wind and solar production by season, Spain 2025. Red bands: critical hours; blue bands: flat hours. Solar production concentrates in midday hours (11–14, white band between the two red bands) — precisely the hours we exclude from the control. The high σ_{within} in critical hours coincides with the morning solar rise and the evening solar fade (see Table 2 for the per-hour values); the within-hour residual-demand swings in those windows are driven jointly by demand ramps and solar transitions. Reproduced from main paper §2.1.

8 Who price-sets

8.1 For the day-ahead market

Method (compressed). OMIE submits stepwise orders to EUPHEMIA: in-the-money fully accepted, out-of-the-money fully rejected, **at-the-money can be curtailed**. The price-setter is the unit whose at-MCP step is partially accepted. Operationally, with q_{below} = sum of unit’s bids strictly below MCP, q_{at} = sum at MCP (± 0.01 EUR/MWh), and q_{assigned} = unit’s PDBC cleared MWh:

$$q_{\text{below}} + \varepsilon < q_{\text{assigned}} < q_{\text{below}} + q_{\text{at}} - \varepsilon$$

Excluded: full acceptance ($q_{\text{assigned}} \geq q_{\text{below}} + q_{\text{at}}$, inframarginal at MCP) and curtailed-to-zero ($q_{\text{assigned}} \leq q_{\text{below}}$, often MIC deactivation). Sources: clearing price from OMIE day-ahead clearing-price series; bid stack from OMIE day-ahead bid-detail and bid-header tables; cleared MWh from OMIE day-ahead programs. **Not** PDBF (bilaterals) or PHF (REE redispatch). Aggregations below MW-weighted by q_{at} unless stated. Paradoxical rejection only applies to block orders (not in DET), so no separate filter is needed.

Naive vs corrected. A naive rule that flags every unit with $p_{\text{bid}} = \text{MCP}$ (this is the rule CNMC uses in its annual market-supervision report, IS/DE/013/24; see Table 5 below) would pool all three cases; the direction of bias is tech-specific. For CCGT 2025Q4 critical, the CNMC-style 9.1% becomes 2.7% under partial-acceptance (most CCGT at-MCP steps are MIC-deactivated to zero); for 2024Q4 critical, 13.8% becomes 18.1%.

Side composition: sell-side dominates critical, buy-side dominates flat/midday in MTU15-DA. Periods where at least one sell unit / at least one buy unit has an at-MCP step.

Regime, hour-class	sell-only	buy-only	both	neither	<i>n</i> periods
DA15/ID15 Critical	53%	25%	11%	11%	4,048
DA15/ID15 Flat	36%	42%	10%	12%	1,104
DA15/ID15 Midday	28%	38%	18%	17%	1,472
DA60/ID15 post-blk Critical	53%	19%	14%	14%	1,716
DA60/ID15 post-blk Flat	60%	17%	9%	13%	468
3-sess Critical	52%	21%	11%	15%	1,870

Table 3: At-MCP side presence. “neither” = EUPHEMIA price-indeterminacy zone (MCP between bid steps, mid-point rule). In DA15/ID15 flat and midday, more periods are buy-side-only price-setting than sell-side-only.

Table 4: Price-setter shares by technology and hour-class, across 5 reform-window regimes (sell-side partial-acceptance, MW-weighted by q_{at}). C = Critical, F = Flat. Caveat: DA60/ID15 pre-blackout is 40 days only — directional.

Tech	3-sess		ISP15-win		DA60/ID15 pre		DA60/ID15 post		DA15/ID15	
	C	F	C	F	C	F	C	F	C	F
Wind	28.3	17.9	37.8	63.9	47.5	70.9	29.6	17.3	29.3	53.0
Solar PV	17.7	0.0	1.4	0.0	20.6	0.0	29.9	0.0	7.7	0.0
Hydro pump	10.4	13.3	17.0	2.7	2.2	10.8	8.0	19.3	28.4	13.0
Hydro	11.9	36.1	19.6	10.9	2.7	1.7	9.4	21.5	23.7	20.0
Nuclear	12.5	20.2	4.2	6.5	15.6	13.5	7.9	5.5	4.4	3.3
CCGT	8.0	8.9	17.2	11.1	0.8	0.0	6.2	33.1	2.7	2.0
Cogen	3.0	0.3	0.1	0.1	2.5	0.1	2.1	0.1	0.5	0.9
Other RES	3.9	1.5	0.6	0.2	3.4	0.8	3.1	1.2	1.2	2.2
Import	0.8	0.8	0.5	3.1	0.0	0.0	0.0	0.0	0.5	3.3

Sell-side price-setter shares by technology and regime (headline table).

Cross-check: CNMC’s broader marginal-tech metric. The Spanish regulator CNMC publishes an analogous metric in its annual market-supervision report (*Informe de supervisión del mercado peninsular mayorista al contado de electricidad. Año 2023*, IS/DE/013/24, 28 Jan 2025), where it states for example that “los ciclos combinados se convierten en la tecnología marginal en un porcentaje elevado de las horas (21,7% cuota ciclos, 43,3% cuota renovable en septiembre” of 2023 (§2.5.3, p. 51). CNMC’s metric counts a unit as marginal whenever its highest-accepted sell tranche lands at MCP, *regardless of whether the at-MCP step is curtailed or fully accepted*. Our partial-acceptance metric above is strictly tighter (it requires the at-MCP step to be *strictly partially* accepted). Table 5 reports the CNMC-style version of the CCGT row of Table 4 for our 5 reform windows; the two metrics give the same qualitative ranking but the CNMC version is mechanically larger.

Table 5: CCGT marginal-tech share, CNMC-style vs partial-acceptance. CNMC-style = % of MW-weighted votes where the rank-1 accepted sell tranche (highest p_{bid} with $p_{\text{bid}} \leq \text{MCP}$) belongs to a CCGT unit; ties at rank 1 split the vote equally across units. This matches the share concept used in CNMC IS/DE/013/24 Cuadro 11 (p. 81). Partial-acceptance = the strict EUPHEMIA-aware rule of Table 4. C = Critical, F = Flat, M = Midday. CNMC’s reported September 2023 figure for CCGT (21.7%, peninsular monthly average) is roughly the order of magnitude of our 3-sess critical-hour CCGT share under the CNMC-style metric (13.1%) and the ISP15-win critical-hour share (14.1%).

Regime	CNMC-style (rank-1 at MCP)			Partial-acceptance (ours)		
	C	F	M	C	F	M
3-sess (Jun–Nov 2024)	13.1	5.9	6.9	8.0	8.9	—
ISP15-win (Dec24–Mar25)	14.1	13.0	14.7	17.2	11.1	—
DA60/ID15 pre-blk (Mar19–Apr27)	2.2	0.0	0.7	0.8	0.0	—
DA60/ID15 post-blk (Apr28–Sep)	11.1	20.5	0.4	6.2	33.1	—
DA15/ID15 (Oct–Dec 2025)	9.1	9.3	1.9	2.7	2.0	—

Why the two metrics diverge. The largest gap is DA15/ID15 critical (9.1 CNMC vs 2.7 ours): under MTU15-DA, most CCGT at-MCP steps are *MIC-deactivated to zero* ($q_{\text{assigned}} \leq q_{\text{below}}$), which the CNMC rule still counts as “marginal” but the partial-acceptance rule does not. The DA60/ID15 post-blackout flat row goes the other way (20.5 CNMC vs 33.1 ours), because in that regime a couple of Endesa CCGT units (SROQ2, PGR5) submit single-block full-nameplate tranches at MCP that get *partially* accepted (driving up the partial-acceptance share) but are not the rank-1 marginal step in many of those periods. Both metrics agree that DA60/ID15 pre-blackout has near-zero CCGT marginal presence and that DA15/ID15 critical is much lower than ISP15-win or 3-sess.

CCGT firm-level: GN never price-sets, IB and GE swap dominance. Within the CCGT column above, who owns the at-MCP step?

Regime	Hour-class	IB	GE	GN	HC	OTHER
3-sess	Critical	60	31	0	0	8
3-sess	Flat	16	50	0	0	25
ISP15-win	Critical	35	57	0	0	5
ISP15-win	Flat	2	74	0	3	20
DA60/ID15 post-blk	Critical	31	69	0	0	0
DA60/ID15 post-blk	Flat	19	80	0	0	0
DA15/ID15	Critical	65	29	0	6	0
DA15/ID15	Flat	80	0	6	14	0

Table 6: Per-firm shares within the CCGT price-setter pool (% of CCGT q_{at}). GN’s CCGT bid stack is so heavily concentrated at scarcity prices (≥ 500 EUR/MWh) that GN essentially never lands at MCP. IB and GE alternate as the dominant CCGT price-setter. The 33.1% CCGT flat-hour figure in DA60/ID15 post-blackout is 80% GE.

Top price-setting units in DA15/ID15 (for context). MUEL (Iberdrola’s Muela del Cortes pump-storage) is the workhorse: 496 events / ~ 5.5 events per day. Other Iberdrola hydro (DUER 201, TAJO 171, SIL 130 events) and the IBEVD11 wind aggregator (153 events, but $\sim 1,613$ MW per event = 247 GW total q_{at}) dominate the top 7. Endesa’s MLTG pump-storage (164 events) is the only non-Iberdrola unit in the top 5.

Reading across regimes (synthesis).

- **Wind is the largest sell-side price-setter** in almost every cell (18–71% MW share). RE-Mercado aggregator portfolios systematically land at-the-money; 70–99% of wind/solar/cogen at-MCP events involve a single block at MCP with no in-the-money bids (the deep-dive companion data). CCGT, hydro, and pump-storage instead use true stacks (24–34% single-block).
- **Iberdrola dominates the marginal MW in DA15/ID15** (top-1 firm-share $\approx 65\%$ critical, 80% flat; firm HHI $> 4,000$), through Hydro pump (MUEL), Hydro (DUER, TAJO, SIL), and the IBEVD11 wind aggregator.
- **CCGT’s share fluctuates strongly across regimes:** 8% (3-sess) $\rightarrow$ 17% (ISP15-win) $\rightarrow$ 0.8% (DA60/ID15 pre-blk, tiny window) $\rightarrow$ 6% / 33% (flat) (DA60/ID15 post-blk) $\rightarrow$ 3% (DA15/ID15) in critical hours. The 33% flat-hour spike is 80% GE-owned, driven by ~ 2 units (SROQ2, PGR5 — both Endesa) submitting single full-nameplate (~ 360 MW) tranches at MCP (MIC-reservation style); cell-count-weighted alternative is 17%.
- **GN does not partial-accept at MCP, but is strategically active within-hour.** GN’s CCGT bid mass sits at scarcity prices (≥ 500 EUR/MWh) and rarely lands at MCP for partial acceptance, but the bid *shape* across the 4 within-hour quarters is the largest of any CCGT firm in terms of median price shift (see Table 13 below). Direct inspection of BES4 (GN) confirms: on 2025-10-15 hour 18, Q1 is binary (1 EUR / 1000 EUR), Q2 the same, Q3 adds a small near-clearing tier, Q4 has a full 10-step ladder at 81–97 EUR/MWh.
- **Solar PV** dominates midday post-MTU15-DA (35.1%, see CSV) and the negative/zero-MCP scarcity bin (40–65%). CCGT never sets the price at $\text{MCP} \leq 0$.
- **Buy-side price-setters** are dominated by pump-load demand bids (46–84%) and retailer demand bids (14–53%). In DA15/ID15 flat hours, buy-side sets the price about 42% of periods (vs 36% sell-side).
- **Cross-regime CCGT symmetry** (simple TTF/gas-shock falsifier) holds in 3-sess (8.0 vs 8.9) and DA15/ID15 (2.7 vs 2.0), but breaks in ISP15-win (17.2 vs 11.1) and DA60/ID15 post-blk (6.2 vs 33.1).

8.2 For the intraday markets

Method. Identical partial-acceptance rule as §8.1, with three data substitutions: clearing price from OMIE intraday-auction clearing-price series (per session $\times$ period), bid stack from OMIE intraday-auction bid-detail data + OMIE intraday-auction bid-header data (sell-side, latest version, pools across the 3 IDA sessions per day), cleared MWh per unit from OMIE intraday-auction programs per unit. Matched-period share is 26–32% across regimes (similar to DA).

Table 7: IDA price-setter shares by technology, hour-class, and regime (at-the-money partial-acceptance, sell side), MW-weighted by q_{at} . C = Critical, F = Flat. Shares sum to 100 per (regime, hour-class).

Tech	3-sess		ISP15-win		DA60/ID15 pre		DA60/ID15 post		DA15/ID15	
	C	F	C	F	C	F	C	F	C	F
Wind	35.5	15.7	24.8	40.2	52.6	53.8	37.8	18.8	31.9	42.4
CCGT	14.1	29.2	16.1	15.3	3.1	2.8	9.7	23.1	24.2	32.6
Hydro pump	12.9	23.4	17.2	2.2	2.5	13.0	6.2	26.1	14.0	3.8
Solar PV	10.5	0.0	1.7	0.1	20.3	0.0	22.8	0.0	7.5	0.1
Hydro	6.4	15.9	18.4	23.4	4.4	3.4	5.6	15.6	10.0	10.6
Retailer	10.2	6.2	10.4	12.1	5.3	7.5	7.3	11.8	5.7	5.2
Nuclear	2.7	3.2	0.0	0.0	1.0	13.8	2.5	1.3	2.5	0.0
Other RES	1.0	1.1	1.7	0.0	3.1	0.7	2.4	0.2	0.6	0.4

Reading the IDA table.

- **CCGT is much more active as an IDA price-setter than as a DA price-setter in DA15/ID15.** Critical-hour CCGT IDA share is **24.2%** (vs **2.7%** in DA); flat-hour CCGT IDA share is **32.6%** (vs **2.0%** in DA). A factor of ~ 9 –16. The natural reading: CCGTs sit out of DA (bidding above the clearing price; see also the q_1 -drop story in Part B, §11) and re-enter via IDA, where they now systematically clear at the marginal step.
- **Wind dominates IDA too**, but less than in DA: 32% IDA critical share in DA15/ID15 vs 29% DA critical share. Wind aggregators play in both markets.
- **CCGT IDA share collapsed in the DA60/ID15 pre-blackout window** (3.1% critical) but recovered post-blackout (9.7%) and grew to 24.2% under DA15/ID15. The post-MTU15-DA jump is striking and consistent with a DA-to-IDA strategic substitution.
- **Retailer (buy-side, IDA)** sets the price 5–12% of IDA critical-hour mass — comparable order to DA. Buy-side matters in IDA too.

9 Bid-curve shape: per-curve functionals and pairwise divergences

We measure bid-curve shape with two conceptually distinct families of statistics, and keep them separate throughout.

- **Family 1 — per-curve functionals.** A functional maps a *single* bid curve to a scalar, $T : f \mapsto T(f) \in \mathbb{R}$; no comparison between curves enters its definition. Cross-regime, cross-firm and cross-hour reading is then done by comparing the *values* $T(f)$ across curves. This family holds the price spread σ_p and the effective tranche count N_{eff} (§9.1), and the functional-PCA level/tilt scores ξ_{ik} of §9.5 — the score $\xi_{ik} = \langle f_i - \bar{f}, \phi_k \rangle$ is itself a per-curve functional, a shape coordinate of the single curve f_i . **This is the primary descriptive object:** it asks what one bid curve looks like, and how that value moves across regimes.
- **Family 2 — pairwise divergences.** A divergence is a statistic of *two* curves, $D(f_i, f_j)$: it does not summarise either curve on its own, it measures how far apart they are. This family holds D_w and its $\Delta p / \Delta q$ channel decomposition (§9.2). We use it for one comparison only — a unit’s four within-hour quarter curves against each other — as a within-hour-shaping diagnostic, secondary to Family 1.

The distinction is the one the advisor asked for: characterise each bid curve by its own value (Family 1), rather than by divergences between curves (Family 2). Family 2 is retained only because the within-hour quarter comparison has no single-curve analogue.

9.1 Family 1 — per-curve functionals: σ_p and N_{eff}

The strategic band, chosen data-driven. Both families restrict to a price band $|p - \text{MCP}| \leq h$ — the region where a bid is in contention — and we read h off the data rather than assert it. Figure 4 plots the MW-weighted density of the bid-to-clearing distance $|p_{\text{bid}} - \text{MCP}|$ over all day-ahead sell tranches of the price-setting techs. It is **bimodal**: a dense near-MCP *competing* cluster and a sparse far *scarcity-cap* cluster (withholding bids near the price cap). We set h at the **antimode** — the first local minimum past the competing cluster. That cluster ends in a sharp cliff near 130 EUR/MWh and the valley bottoms at $h = 230$; this band holds 62% of sell-bid MW and excludes the cap cluster by construction, where the legacy $h = 50$ kept only $\sim 26\%$ of CCGT sell-bid MW and understated every shape statistic. h is estimated once — pooled across regimes and techs — and frozen; a regime- or tech-specific band would move with behaviour and contaminate the cross-comparisons.

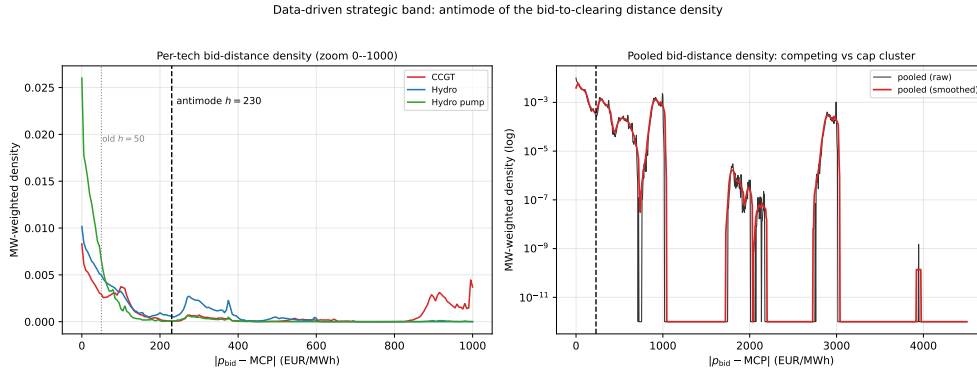


Figure 4: Data-driven strategic band. MW-weighted density of $|p_{\text{bid}} - \text{MCP}|$ for day-ahead sell tranches (price-setting techs, all regimes pooled). Left: per-tech, zoomed to 0–1000 EUR/MWh. Right: pooled, log scale — the near-MCP competing cluster and the scarcity-cap cluster show as two modes. The band edge $h = 230$ (dashed) sits at the antimode between them; the legacy $h = 50$ (dotted) fell deep inside the competing cluster.

Price spread σ_p and effective tranche count N_{eff} . For each (unit, date, market-period) sell bid curve, restricted to the band $|p - \text{MCP}| \leq h$, we compute two scalars:

- **Price functional — price spread σ_p :** the MW-weighted standard deviation of the in-band tranche prices (EUR/MWh). It measures how much the unit *varies its price* across the MW it offers near the clearing price. A single flat block $\Rightarrow \sigma_p = 0$; a graded ladder spanning the band $\Rightarrow \sigma_p$ large.
- **Quantity functional — effective tranche count N_{eff} :** the inverse Herfindahl of the tranche MW shares, $N_{\text{eff}} = (\sum_t q_t)^2 / \sum_t q_t^2$. It measures how the unit *splits its quantity*: $N_{\text{eff}} = 1$ means all in-band MW sits in one block, larger N_{eff} means the MW is divided into more comparable pieces.

Both are per-curve and require no curve-to-curve comparison, and are computed on the *raw* bid curve (seasonality is handled at the aggregation step, by reading the values within calendar-matched windows). The two decouple: a unit can split into many tranches at nearly one price (high N_{eff} , low σ_p) or place two blocks far apart (low N_{eff} , high σ_p). The fPCA scores of §9.5 are the third member of this family — a multivariate per-curve functional that lets the data choose the modes of shape variation rather than fixing them a priori.

9.2 Family 2 — pairwise divergences: D_w , Δp , Δq

D_w , in full. Let cell (f, u, d, h) index a firm f , unit u , date d , clock-hour h . The hour partitions into 4 quarters $q \in \{1, 2, 3, 4\}$. In quarter q the unit submits a multiset of price-quantity sell

tranches with $p \in [\underline{p}, \bar{p}] \subset \mathbb{R}$ (Spanish SDAC: $\underline{p} = -500$, $\bar{p} = 4,000$ EUR/MWh; negative bids are admissible) and $x > 0$ (MWh, energy for the quarter-slot). Let $c_{d,h}$ be the hour-mean clearing price (justified below) and adopt the data-driven band half-width h of §9.1 (the antimode of the bid-distance density, $h = 230$ EUR/MWh) around it. **We work throughout with band-restricted supply curves:** N_q is the number of *distinct* prices submitted in quarter q *within* $[c - h, c + h]$, and x_{qk} is the sum of quantities at p_{qk} . The band-local quarter- q supply curve is the right-continuous step function

$$S_q(p) = \sum_{k=1}^{N_q} x_{qk} \mathbf{1}\{p_{qk} \leq p\}, \quad p \in [c - h, c + h],$$

with $S_q(c - h) = 0$ and $S_q(c + h) = M_q$ (the band-mass of quarter q — the total MWh offered *anywhere inside* the band $[c - h, c + h]$, not the offer at the maximum-price threshold). Infra-marginal mass below $c - h$ is excluded by construction: D_w measures the within-band *shape* of supply across quarters, not infra-marginal level offsets.

For two quarters i, j , the 1-D Wasserstein-1 distance is the L^1 area between cumulative curves, equivalently the L^1 distance between (extended) quantile functions:

$$W_1^{\text{band}}(S_i, S_j) = \int_{c-h}^{c+h} |S_i(p) - S_j(p)| dp = \int_0^{\max(M_i, M_j)} |S_i^{-1}(Q) - S_j^{-1}(Q)| dQ,$$

where $S_q^{-1}(Q) = c + h$ for $Q > M_q$ pads the smaller curve to common total mass (Vallender, 1973). To isolate variation near the clearing price, weight by an (unnormalized) Epanechnikov-shape kernel:

$$K_h(p - c) = \max\left\{0, 1 - \left(\frac{p - c}{h}\right)^2\right\}, \quad K_h \in [0, 1], \quad K_h(c) = 1,$$

$$D_w^{(i,j)} = \int_{c-h}^{c+h} K_h(p - c) |S_i(p) - S_j(p)| dp \leq W_1^{\text{band}}.$$

D_w has units EUR (quantity in MWh times price in EUR/MWh).

Why $c = \text{hour-mean clearing price}$. The band isolates the strategic pricing region — the range where the unit's bids had a non-trivial probability of clearing. We centre on $c_{d,h}$, the hour-mean of the four within-hour quarter clearing prices, so the band is common to all 6 quarter-pairs in a cell; a quarter-specific centre would measure shifts of the reference, not of the curve.

Band. D_w , Δp and Δq use the same data-driven strategic band as the per-curve functionals (§9.1). *Pending refresh:* the D_w / Δp / Δq tables in §9.4 were computed at the legacy fixed $h = 50$ and have not yet been re-run at the data-driven $h = 230$. The refresh is held as a separate analytical step, not a mechanical re-run: a band $\sim 4.6\times$ wider can saturate the binary $D_w > 0$ flag (almost every cell acquires *some* within-hour quarter difference), which would change what the diagnostic discriminates — a substantive choice for the pairwise family, distinct from the per-curve functionals where a wider band simply admits more of the genuine competing cluster.

Cell-level aggregation: mean across the 6 within-hour pairs. A cell has $\binom{4}{2} = 6$ pairwise dissimilarities $D_w^{(i,j)}$, aggregated to a cell-level statistic by the *mean* $D_w^{\text{cell}} = \frac{1}{6} \sum_{i < j} D_w^{(i,j)}$. For the binary flag $D_w^{\text{cell}} > 0$ the choice of mean vs. max is irrelevant (a cell has any pair > 0 iff mean > 0 iff max > 0); for conditional intensity we report the mean, which reads as the typical-pair within-hour shift.

Decomposition into price and quantity channels. We split the within-band variation into a price channel and a quantity channel:

$$\Delta q_{ij} = |M_i - M_j| \text{ [MWh]}, \quad \Delta p_{ij} = \frac{1}{\min(M_i, M_j)} \int_0^{\min(M_i, M_j)} |S_i^{-1}(Q) - S_j^{-1}(Q)| dQ \text{ [EUR/MWh]}.$$

Δp captures the mean price shift on the common-mass portion — a price-only adjustment with band-mass totals held fixed, in EUR/MWh. Δq captures band-mass mismatch — a quantity-only adjustment with prices on the common-mass portion held fixed, in MWh. Cell-level Δp , Δq are the mean over the 6 pairs, consistent with the D_w aggregation above.

Consistency and the Δq caveat. Extending the smaller quantile function to common mass M_j and splitting the Wasserstein integral at $Q = M_i$ gives the bound

$$D_w^{(i,j)} \leq W_1^{\text{band}} \leq \Delta p_{ij} M_i + 2h \Delta q_{ij},$$

exact in the two limiting cases: $M_i = M_j \Rightarrow$ all variation on the price channel; coinciding quantile functions on common mass $\Rightarrow$ all variation on the quantity channel. **Read Δp and Δq as channel indicators (active/inactive, and which dominates), not as additive contributions** — empirically the bound is loose by $\sim 13\times$ (the extra band-mass sits near the top of the band). Raw Δq in MWh is also coarse on its own: 10 MWh of asymmetric mass is large in a 20 MWh cell and negligible in a 500 MWh one, so we lean on the active/inactive distinction rather than the raw level.

9.3 Bid graphs by tech, firm, and within-hour quarter

The figures in this subsection display the aggregate within-hour quarter bid (or demand) curves, broken down by firm and hour-class. Each curve overlays the mean DA (or IDA) clearing price as a reference so the strategic-band $|p - c| \leq 230$ EUR/MWh region is visible. Aggregation: average across ~ 92 dates per (firm, hour, quarter) cell for the DA Oct–Dec 2025 window. Day-specific quarter dispersion is smoothed by this construction; the per-curve price and quantity metrics in §9.4 are the right quantitative complement.

CCGT (DA): aggregate per-firm quarter curves, critical vs flat. *Reading (mixed).* The aggregate construction smooths per-day strategic shaping; the per-curve metric distributions at the end of §9.4 are the right systematic measure.

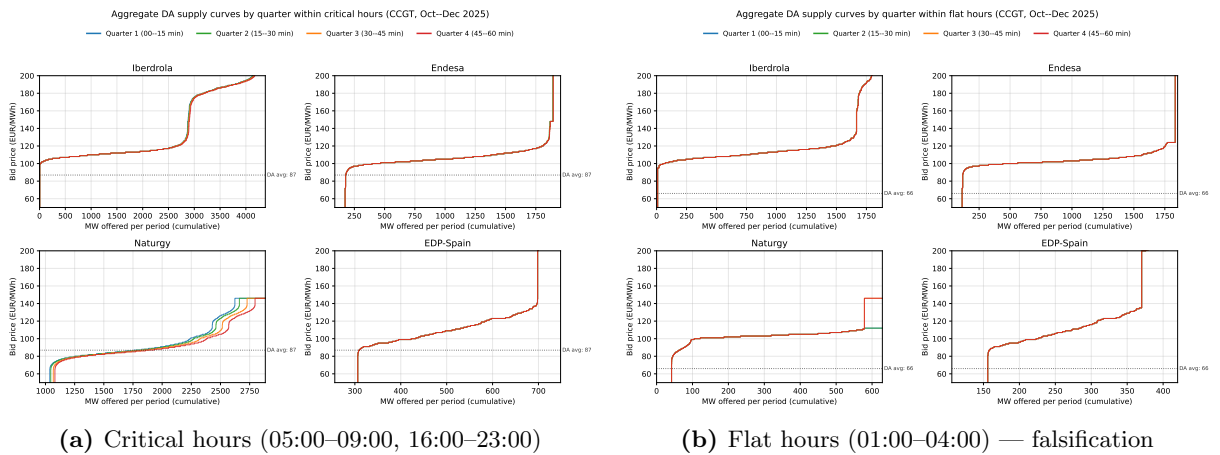


Figure 5: Aggregate DA supply curves by quarter, CCGT, Oct–Dec 2025. Aggregated across ~ 92 dates per (firm, hour, quarter) cell; day-specific quarter dispersion is smoothed in this visual.

In the post-MTU15-DA window the IDA per-firm pattern matches DA (dispersion concentrated in Naturgy, visible only in critical hours); the IDA-side figure is not reproduced separately.

Compact multi-tech grids: firm $\times$ tech $\times$ quarter, all techs side by side. The grids below stack rows = Big-4 firms (Iberdrola, Endesa, Naturgy, EDP-Spain) against columns = techs (CCGT, Hydro, Nuclear, Wind, Solar PV). Each panel overlays the 4 within-hour quarter supply curves (Q1 blue $\rightarrow$ Q4 red) with the mean DA / IDA clearing price as a dotted horizontal line. *Reading:* the same-quarter overlap in flat hours is the falsification; visible quarter spread in critical hours is the strategic signal.

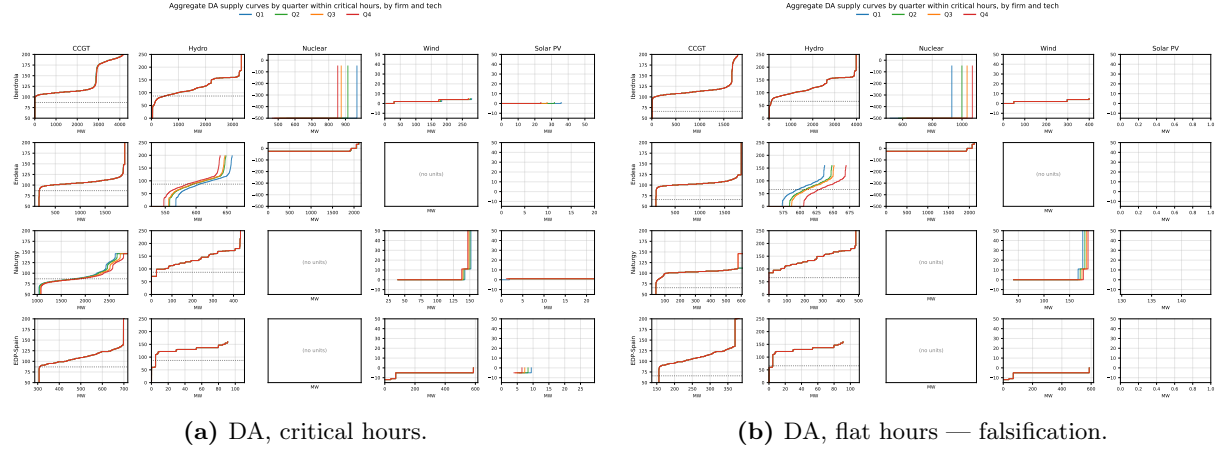


Figure 6: DA compact grid (firms $\times$ techs $\times$ quarter), DA15/ID15. Quarter overlap in flat hours is essentially complete; critical-hour spread is visible for GN CCGT and Endesa Hydro.

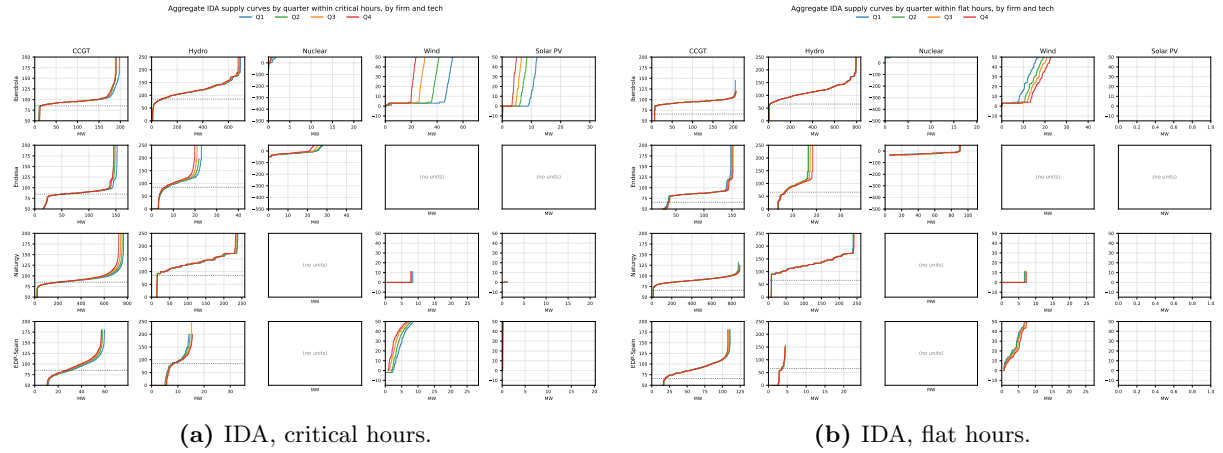


Figure 7: IDA compact grid (firms $\times$ techs $\times$ quarter), DA15/ID15. IDA-side analogue; quarter spread on the supply side is more visible than in DA.

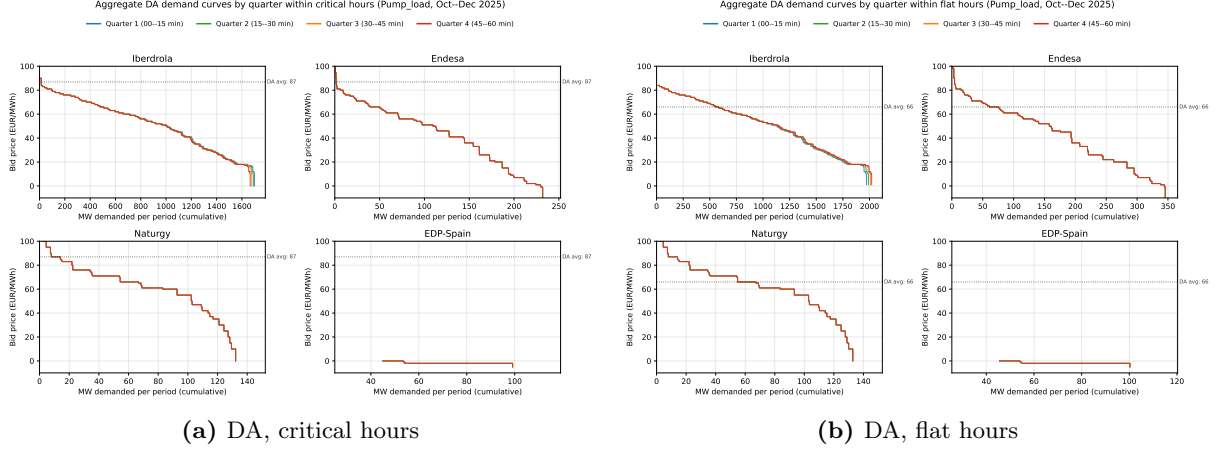


Figure 8: Aggregate DA demand curves by quarter, pump-storage charge bids, Oct-Dec 2025. The IDA-side analogue is similar and is not reproduced.

9.4 Results: per-curve functionals, and the pairwise D_w diagnostic

The results below quantify what the bid-curve figures in §9.3 show visually: first the Family-1 per-curve functionals (σ_p , N_{eff} , defined in §9.1), then the Family-2 pairwise D_w diagnostic (§9.2).

Per-curve functionals: price spread and effective tranche count. Table 8 and Figure 9 report σ_p and N_{eff} averaged over (unit, date, period) day-ahead sell-bid curves, by tech, hour-class and regime — the cross-regime reading is a comparison of per-curve *values*, with no differencing of curves.

Table 8: Per-curve bid-shape metrics by tech, hour-class and regime (DA sell bids, in-band $|p - \text{MCP}| \leq 230$ — the data-driven band of §9.1 — mean over (unit, date, period) curves). σ_p = MW-weighted price spread (EUR/MWh); N_{eff} = effective in-band tranche count. 994,616 in-band curves.

Tech	Hour	3-sess		ISP15-win		DA60/ID15 pre		DA60/ID15 post		DA15/ID15	
		σ_p	N_{eff}	σ_p	N_{eff}	σ_p	N_{eff}	σ_p	N_{eff}	σ_p	N_{eff}
CCGT	Critical	40.5	3.2	51.4	3.3	35.1	3.3	28.0	3.1	28.6	3.4
CCGT	Flat	27.3	2.6	44.8	2.7	21.3	2.4	20.2	2.6	19.1	2.3
Hydro	Critical	18.2	3.6	18.8	3.5	16.1	3.6	18.4	3.5	21.6	3.8
Hydro	Flat	18.0	3.6	18.3	3.4	16.4	3.5	19.0	3.5	20.1	4.0
Hydro pump	Critical	3.8	2.4	4.0	2.4	3.5	2.3	4.3	2.8	4.7	3.3
Hydro pump	Flat	2.1	2.5	3.8	2.8	3.7	2.3	2.8	2.5	2.9	3.8

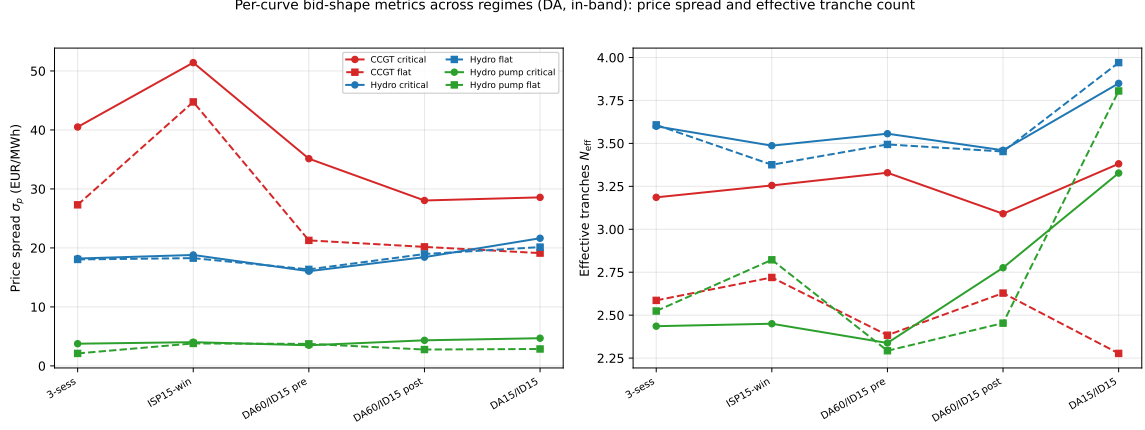


Figure 9: Per-curve bid-shape metrics across regimes (DA, in-band, data-driven band $|p - \text{MCP}| \leq 230$). Left: price spread σ_p ; right: effective tranche count N_{eff} . Each line is a (tech, hour-class) cell; solid = critical hours, dashed = flat. CCGT carries the widest price spread ($\sigma_p \approx 28\text{--}51$ EUR/MWh) and it is consistently larger in critical than flat hours in every regime — the strategic signal; Hydro is intermediate ($\sim 16\text{--}22$) and Hydro pump near-flat (~ 4). N_{eff} sits at $\sim 3\text{--}4$ for the price-setting techs, rising mildly for Hydro pump across the reform sequence.

The pairwise D_w diagnostic (Family 2). The $D_w / \Delta p / \Delta q$ statistics below are the Family-2 *pairwise* construction (§9.2) — they compare a unit’s four within-hour quarter curves to each other. They remain a useful within-hour-shaping diagnostic, but the per-curve functionals above are the primary descriptive object. We report (i) the per-cell binary flag $D_w > 0$ across techs and firms (operational vs strategic decomposition, IDA pre/post-blackout) and (ii) the $\Delta p / \Delta q$ channel-shares and intensity tables.

DA Oct–Dec 2025: operational vs strategic decomposition by (tech, firm). *Reading (mixed).* The original main-paper framing reported a critical–flat asymmetry with the interpretation “only Naturgy CCGT shows the strategic signature”. The IDA pre/post-blackout analysis (below) and the multitech within-hour shaping table (at the end of this subsection) complicate this: GN’s CCGT D_w activity is real and large, IB hydro and pump-storage are also highly active, and under reforzada all four firms show the IDA-side strategic signature, not just Naturgy.

Table 9: DA Oct–Dec 2025 critical–flat asymmetry, by (channel, technology, firm). Percentage of (firm, unit, date, hour) cells with $D_w > 0$. *Channel:* Strategic = exploitation of within-hour granularity near clearing; Operational = forecast/load-shape updates; Mechanical = smooth pump-storage charge. Compare: IDA in Table 10, multitech in Table 13.

Channel	Tech	Firm	% crit	% flat	Crit–Flat (pp)
Strategic	CCGT	GN	24.5	0.0	+24.5
Strategic	CCGT	IB	5.8	3.9	+1.9
Operational	Wind	IB	97.2	97.3	−0.1
Operational	Wind	GN	92.7	92.6	+0.1
Operational	Wind	HC	93.7	93.8	−0.1
Operational	Solar PV	GN	15.9	0.0	+15.9
Operational	Retailer	GE	76.5	76.3	+0.2
Operational	Retailer	IB	40.9	40.2	+0.7
Operational	Direct_consumer	IB	96.0	95.3	+0.7
Mechanical	Pump_load	IB	3.2	5.5	−2.3
Mechanical	Pump_load	GE	0.0	0.0	0.0
Mechanical	Hydro_pump	IB	0.4	0.0	+0.4
Mechanical	Hydro_pump	GE	0.0	0.0	0.0

IDA pre/post-blackout near-clearing dissimilarity. For each (firm, unit, date, hour) CCGT IDA cell, Epanechnikov-weighted within-hour quarter dissimilarity D_w ($h = 50$ EUR/MWh, centred on hour-mean IDA clearing price). Pre-blackout: 2025-03-19 $\rightarrow$ 2025-04-27. Post-blackout: 2025-04-28 $\rightarrow$ 2025-09-30 (reforzada). Hypothesis: REE post-clearing intervention should remove the leverage on clearing price, so pre $>$ post.

Table 10: IDA quarter-dissimilarity, pre vs post-blackout, CCGT. Percentage of (firm, unit, date, hour) cells with $D_w > 0$ (i.e., quarters differ within the Epanechnikov band, $h = 50$ EUR/MWh, around hour-mean IDA clearing price). *Pre-blackout*: 2025-03-19 $\rightarrow$ 2025-04-27 (40 days, MTU15-IDA active, reforzada-free). *Post-blackout*: 2025-04-28 $\rightarrow$ 2025-09-30 (155 days, reforzada active).

Firm	Pre-blackout			Post-blackout		
	Critical	Flat	Crit–Flat	Critical	Flat	Crit–Flat
IB	9.6	7.0	+2.6	20.3	6.2	+14.1
GE	14.1	10.0	+4.1	43.8	33.2	+10.6
GN	30.5	9.2	+ 21.3	53.8	37.9	+15.9
HC	27.9	20.7	+7.2	34.2	21.2	+13.0

Reading (mixed). (i) Pre: only GN shows clean asymmetry (+21pp). (ii) Post: all four firms show +11 to +16pp gap. (iii) Pre $>$ post rejected for every firm; rates roughly double. (iv) Flat-hour rates post-blackout 6–38% (reforzada-driven operational adjustments); the critical-flat asymmetry is the *lower bound* on strategic activity.

Verdict (mixed). The directional prediction (pre $>$ post) is rejected, but the critical-vs-flat asymmetry survives and broadens to all four firms. Reading: pay-as-bid RT2 + intra-tech competition $\Rightarrow$ the strategic-bidding mechanism is essentially unchanged from pre-reforzada. The relevant new lever is geographic (§13).

$\Delta p / \Delta q$ channel shares (CCGT, DA Oct–Dec 2025). *Methodology and decomposition:* see §9.2. *Outcomes below are the share of cells where each channel is active and the median value conditional on active. Caveat: raw Δq in MWh is coarse on its own — see the Δq caveat in §9.2.*

Table 11: DA Oct–Dec 2025, CCGT. Outcomes: Δp -active (%) = % of cells with $\Delta p_{ij} > 0$ for at least one within-hour pair (price channel is active); Δq -active (%) = same for the quantity channel; *med Δp* = median pair-wise Δp in EUR/MWh over the cells where the price channel is active; *med Δq* = same in MWh for the quantity channel. Strategic band $|p - c| \leq 50$ EUR/MWh. Pair-wise statistics aggregated to cell via max across the 6 quarter-pairs.

Firm	Hour	n cells	Δp -active (%)	Δq -active (%)	med Δp (EUR/MWh)	med Δq (MWh)
GN	critical	14 930	0.01	18.4	3.4	147
GN	flat	4 065	0.00	0.00	—	—
GE	critical	5 195	1.56	0.33	4.6	364
GE	flat	1 413	0.00	0.00	—	—
IB	critical	7 831	0.10	1.94	3.9	70
IB	flat	2 139	0.89	1.03	3.1	11
HC	critical	2 013	0.00	0.05	—	60
HC	flat	549	0.00	0.00	—	—

Reading (supportive). (i) GN: 18.4% Δq -active, 0.01% Δp -active — shaping is quantity-only. (ii) Flat hours: 0% on both channels for GN/GE/HC. (iii) GE has the opposite mix (1.56%

Δp vs $0.33\% \Delta q$), both small. (iv) IB/HC: negligible. *Caveat*: the raw Δq medians (e.g. GN crit 147 MWh, GE crit 364 MWh) should be read alongside firm-level mean band-mass to interpret the proportional size — see setup section’s caveat on Δq .

Verdict (supportive). GN fixes the price ladder near the cap and re-allocates quantity across the 4 quarters: the Cournot version of price-raising. Unit tests (identical-curves $\rightarrow 0/0$; same-price-diff-mass $\rightarrow 0/+$; same-mass-diff-price $\rightarrow +/0$) confirm the decomposition is mathematically correct.

D_w distributions across firms, technologies, and regimes. The next table summarises the D_w distribution by sample: the mass at $D_w = 0$ (no within-hour quarter variation in the band) and, conditional on $D_w > 0$, the median value.

Table 12: CCGT mass at $D_w = 0$ and conditional median D_w , by sample. Outcome: share of (firm, unit, date, hour) cells with $D_w = 0$ (no within-hour quarter variation in the Epanechnikov band $|p - c| \leq h$ around clearing) and, conditional on $D_w > 0$, the median D_w value (units: EUR — D_w multiplies a quantity in MWh by a price in EUR/MWh, so it has dimension EUR; see §9). Three samples: DA Oct–Dec 2025 (post-MTU15-DA), IDA pre-blackout (2025-03-19 to 2025-04-27), IDA post-blackout (2025-04-28 to 2025-09-30).

Sample	Firm	% crit cells $D_w=0$	% flat cells $D_w=0$	med D_w crit (EUR)	med D_w flat (EUR)
DA Oct–Dec 2025	IB	98	98	415	154
	GE	98	100	1 333	—
	GN	77	100	5 167	2 287
	HC	100	100	—	—
IDA pre-blackout	IB	90	93	142	6
	GE	86	90	503	108
	GN	70	91	504	242
	HC	72	79	735	761
IDA post-blackout	IB	80	94	229	342
	GE	56	67	498	165
	GN	46	62	836	322
	HC	66	79	835	674

Reading (supportive). (i) DA mass-at-zero $\approx$ identical critical vs flat for every firm except GN (77% vs 100%). (ii) IDA: post-blackout shifts mass-at-zero down for every firm; %crit-at-0 < %flat-at-0 in every (firm, regime). (iii) Conditional median D_w in critical hours $\approx 2\text{--}3\times$ flat for GN, GE on IDA. (iv) Strategic signature (mass-asymmetry critical < flat) appears only in CCGT and Solar PV; operational categories overlap.

Within-hour strategic shaping by (tech, firm), DA15/ID15 critical hours. The partial-acceptance frame in §8.1 ranks who clears at MCP. The bid-shape frame here asks who actively shapes within-hour bids across the 4 quarters, regardless of whether their bid lands at MCP. We compute $\Delta p_{\max}$ and $\Delta q_{\max}$ (kernel band ± 50 around MCP). A unit is “ Δp -active” in an hour-cell iff $\Delta p_{\max} > 0$ (similarly Δq -active).

Table 13: Within-hour strategic shaping by tech and firm, DA15/ID15 critical hours. %-act = share of hour-cells with $\Delta_{\max} > 0$; med- Δ conditional on active.

Tech	Firm (top units)	n units	n cells	% Δp -act	% Δq -act	med Δp (EUR/MWh)	med Δq (MWh)
CCGT	IB	4	1,935	68.3	74.5	0.23	382
CCGT	GN	13	5,906	32.6	52.9	4.14	211
CCGT	GE	4	1,570	8.6	77.5	3.48	371
Wind	GST	2	1,076	96.3	98.9	5.06	19
Wind	IB (IBEVD11)	1	250	23.6	27.2	0.34	1,291
Wind	HC (EDP)	3	1,638	9.3	17.2	0.65	238
Wind	AXPO, NEXUS	3	1,013	0.0	13–16	—	1–13
Hydro	IB	3	1,638	73.8	79.4	4.02	448
Hydro	GE (GDLQ)	1	402	22.4	23.9	0.16	73
Pump-st.	IB (MUEL)	1	540	88.0	88.9	1.90	1,221
Pump-st.	GE (MLTG)	1	418	86.1	91.6	3.44	195

Reading (supportive). GN ranks at 0% of CCGT *price-setting* (their bids are too high to land at MCP, see §8.1) but here actively shapes within-hour bids in 33% of critical hours with the largest median Δp of any CCGT firm. Iberdrola is the dominant strategic actor across CCGT (68%), hydro (74%), and pump-storage (88%). Wind aggregators split sharply: Gesternova (independent) ladder-bids across quarters (96% Δp -active); IBEVD11 (Iberdrola, volume giant) shifts quantities but not prices — forecast-driven, not strategic; AXPO and NEXUS bid identically across all 4 quarters. The MTU15-DA reform is exploited by specific firm-unit combinations, not uniformly.

9.5 Functional PCA of bid-curve shape changes across regimes

Motivation and method. The two per-curve functionals of §9.1 (σ_p , N_{eff}) fix the shape coordinate in advance. Functional PCA (fPCA) is the data-driven member of the same Family 1: it still maps each single curve to scalars, but lets the data choose the modes. It treats each per-(unit, date, period) sell bid stack as a functional observation — the inverse supply curve $f_i(q) = S_i^{-1}(q)$ on cumulative-MW quantiles $q \in [0, 1]$ — and extracts orthogonal modes of shape variation. With sample mean $\bar{f}$, fPCA finds an orthonormal basis $\{\phi_k\}$ with uncorrelated scores

$$\xi_{ik} = \langle f_i - \bar{f}, \phi_k \rangle_{L^2} = \int_0^1 [f_i(q) - \bar{f}(q)] \phi_k(q) dq,$$

and the Karhunen-Loève expansion $f_i(q) = \bar{f}(q) + \sum_k \xi_{ik} \phi_k(q)$ replaces each curve by K scalar scores. PC1 typically captures a level shift, PC2 a cheap-vs-scarcity tilt, higher PCs finer shape. Quantile space is scale-normalised across units of any size; we pool per tech (a single pooled PCA would burn PC1 on cross-tech mean differences). To keep PC1 orthogonal to the calendar cycle we deseasonalise functionally before PCA: at each quantile grid point q , subtract the seasonality-only fit of the FWL spec of §, $f_i^{\text{SA}}(q) = f_i(q) - \widehat{\text{seas}}_i(q)$.

Explained variance. The first PC captures most of the variation:

Tech	PC1	PC2	PC3	PC4	PC5	Cumulative top-5
CCGT	0.86	0.08	0.03	0.01	0.01	0.98
Hydro	0.91	0.06	0.01	0.01	0.00	0.99
Hydro pump	0.89	0.07	0.01	0.01	0.00	0.99
Nuclear	0.62	0.17	0.08	0.04	0.02	0.93
Wind	0.78	0.10	0.04	0.02	0.01	0.96
Solar PV	0.89	0.06	0.02	0.01	0.00	0.98
Cogen	0.62	0.19	0.12	0.03	0.02	0.97

Score regression: pairwise reform-window design. Each reform is isolated by restricting the sample to its immediate before/after windows (ISP15: 3-sess $\rightarrow$ ISP15-win; MTU15-IDA:

ISP15-win $\rightarrow$ DA60/ID15 pre-blk; MTU15-DA: DA60/ID15 post-blk $\rightarrow$ DA15/ID15). For each (tech, firm) and reform we run, separately per PC k , on the SA scores:

$$\xi_{i,k}^{\text{SA}} = \alpha + \beta_{0,k} \mathbb{1}[\text{post}_i] + \sum_{h \in \{\text{Flat}, \text{Mid}\}} \beta_{h,k} \mathbb{1}[\text{post}_i] \mathbb{1}[\text{hour}_i = h] + \gamma_k X_i + \varepsilon_{i,k},$$

where X_i collects calendar-month, hour-of-day, and within-hour quarter FE. The post-effect is $\beta_{0,k}$ on Critical hours, $\beta_{0,k} + \beta_{h,k}$ on Flat/Midday. Table 14 reports each pre-post PC change projected back onto the curve domain via $\hat{\beta}_k \cdot \phi_k$ (EUR/MWh, summed across quantiles).

Table 14: Pairwise PC1, PC2, PC3 coefficients across all three reforms (functional-SA), critical hours. Cells in EUR/MWh on the curve domain (post coefficient projected via $\beta_k \phi_k(q)$, summed). The MTU15-IDA panel uses a 40-day post window. RES techs (Wind, Solar PV, Cogen, Solar Thermal) are aggregated per firm into “OTH” (RES has too many small distributed units for per-firm fPCA).

Tech	Firm	ISP15			MTU15-IDA			MTU15-DA		
		PC1	PC2	PC3	PC1	PC2	PC3	PC1	PC2	PC3
CCGT	GE	+3459	+2042	+903	+2560	-260	-1022	-15018	+879	-183
	GN	+315	+109	+32	-206	+107	-38	+862	-736	+30
	HC	-1573	+240	+175	-611	+9	-323	+2480	-270	+166
	IB	-1558	-60	-253	+16	-248	+233	+1849	-583	-218
	AXPO	-1567	-188	-139	+189	+12	+60	+1217	-609	-116
	REP	-1368	+44	-377	-258	-383	+164	+1451	-630	-163
	OTH	-1349	-61	-133	+45	-122	-90	+2183	-485	-78
Hydro	GE	+134	+124	+0	-750	-41	-22	-384	-6	+5
	GN	+351	+95	-8	+252	+58	+31	-173	-59	-42
	HC	+519	-331	+29	-243	-44	+84	+305	-264	-137
	IB	+1047	-134	-108	-269	-91	+112	-188	+25	+6
	REP	-10	-47	-20	-322	+95	+77	+55	-39	-89
	OTH	-374	+100	-8	+85	+30	+12	-702	-18	+3
Hydro pump	GE	-280	+76	-7	-359	-11	+0	-305	-16	+0
	GN	+333	+77	-9	-112	-83	+105	+1272	-19	+3
	IB	-59	-108	-41	-471	-132	+68	-57	-44	-29
	REP	-145	+1	+35	-366	-143	+89	-380	+145	-68
Nuclear	GE	-1076	-361	+64	+0	+512	+238	+1681	-767	-1498
	GN	-632	+152	+79	-65	+396	+140	+2132	-1322	+380
	IB	-842	+87	-13	+62	+393	+188	+763	-1666	-516
Wind	OTH	-50	-9	+4	+17	-1	+1	+8	+17	-12
Solar PV	OTH	+474	+39	+11	+416	-78	+33	+1599	+102	+6
Solar Thermal	OTH	+68	+11	+5	-23	+26	-26	-42	+13	+14
Cogen	OTH	-26	-9	+11	+5	-21	+14	+121	+182	-136

Reading the fPCA evidence.

- **Headline level shifts.** GE-CCGT moves opposite to GN/IB/HC under every reform: up under ISP15 (+3,459 Critical, PC1) and MTU15-IDA, sharply down under MTU15-DA (-15,018 Critical, the largest single SA coefficient); IB-CCGT +1,849, HC-CCGT +2,480, GN-CCGT +862 under MTU15-DA.
- **PC2/PC3 carry weight where the reform changes tilt, not level.** Nuclear under MTU15-IDA has PC1 near zero but PC2 +393 to +512 across GE/GN/IB — a within-curve tilt PC1 misses. CCGT under MTU15-DA has positive PC1 but negative PC2 (-270 to -879) for five of seven firms: cheap-MW tranches repriced up more than the scarcity tail.
- **Identification caveat.** MTU15-DA and reforzada are both active in the MTU15-DA post-window; the coefficient is the joint shift. Only the 40-day MTU15-IDA pairwise is a

clean single-reform read. A year-fold placebo (2022/2023 calendar windows) flags the GE-CCGT ISP15 raw shift as seasonal, while the CCGT IB/HC/GE MTU15-DA critical-hour pattern survives both controls.

10 Continuous-intraday repositioning per technology

Why this matters. The system-cost evidence in §14 documents the redistribution of expenditure between TR (real-time technical restrictions) and Fase I (pre-IDA pay-as-bid redispatch). What the cascade view does not show is the bidding-side mechanism through which the cost reallocation materialises. The hypothesis is that finer-grained intraday clearing absorbs within-hour scarcity that previously spilled into real-time correction; if true, this should be visible in the volume of repositioning that technologies do in the post-IDA continuous market (XBID). We characterise per-technology continuous-intraday (CI) activity across the reform regimes.

Variable. Each XBID matched trade pairs a buyer unit and a seller unit (with a contract quantity in MW for a delivery period of the relevant MTU length). For each (delivery date, technology) we sum the MWh of CI trades on the sell side (the technology of the seller unit) and on the buy side (the technology of the buyer unit), then average per day within each regime. The *gross* CI activity per tech is the sum of sells and buys; the *net* CI position is sells minus buys. **Sources:** XBID matched-trade panel and unit-to-technology bridge.

Table 15: Gross continuous-intraday activity by technology and regime, GWh/day. Sum of sell-side and buy-side MWh per (delivery date, technology), averaged per day per regime. Captures total CI repositioning by tech.

	pre-IDA	3-sess	ISP15-win	MTU15 pre-blk	MTU15 post-blk	DA15/ID15
CCGT	1.3	3.9	4.4	1.9	2.4	3.7
Nuclear	2.4	3.8	3.5	1.3	3.8	4.1
Hydro	3.3	4.2	4.5	2.6	2.4	3.2
Hydro pump	1.6	2.2	2.1	1.5	1.1	1.0
Wind	5.5	6.5	5.1	3.5	3.6	7.5
Solar PV	3.7	3.5	2.1	2.2	3.0	3.5
Solar Thermal	0.4	0.7	0.5	0.6	0.6	0.3
Biomass / RE thermal	0.4	0.6	0.5	0.5	0.4	0.4
Thermal non-RE	0.2	0.3	0.4	0.4	0.3	0.3
Battery / storage	0.0	0.0	0.0	–	0.1	0.1
Cogen	–	–	–	–	–	–

Table 16: Net continuous-intraday position by technology and regime, GWh/day. Sells – buys per (delivery date, technology). Positive values = net seller (technology adds output via CI); negative = net buyer (technology reduces output via CI).

	pre-IDA	3-sess	ISP15-win	MTU15 pre-blk	MTU15 post-blk	DA15/ID15
CCGT	+0.8	+2.5	+2.5	+1.3	+1.6	+2.2
Nuclear	–1.2	–1.0	–1.7	+0.3	–2.9	–0.2
Hydro	+0.7	+0.9	+1.1	+0.2	+0.2	+0.6
Hydro pump	+0.1	–0.0	+0.1	–0.0	+0.1	–0.1
Wind	–0.1	–0.5	–0.4	+0.5	+0.2	–1.2
Solar PV	+0.4	–0.3	–0.1	–0.3	–0.3	+0.1
Solar Thermal	–0.1	–0.1	–0.1	+0.1	–0.2	–0.1
Biomass / RE thermal	–0.1	–0.3	–0.2	–0.2	–0.2	–0.2
Thermal non-RE	–0.0	–0.0	–0.1	+0.0	–0.1	–0.1
Battery / storage	+0.0	+0.0	–0.0	–	+0.0	+0.0
Cogen	–	–	–	–	–	–

Reading.

- **System-wide gross CI activity falls during MTU15-IDA, then recovers under MTU15-DA** ($\sim 20 \rightarrow \sim 13 \rightarrow \sim 19$ GWh/day): the granular IDA absorbs intra-hour

scarcity that previously fell to XBID; the later granular DA produces more per-quarter mismatch for XBID to close.

- **Wind doubles its CI activity under MTU15-DA** ($3.5 \rightarrow 7.5$ GWh/day, the largest tech-specific rise) — quarter-hour forecast-error correction, freed once DA and CI are both granular.
- **CCGTs are systematic net SELLERS in CI in every regime** (+0.8 to +2.5 GWh/day, Table 16): the XBID-side face of the “stay out of DA, re-enter via post-clearing channels” narrative (§11–§13). **Nuclear is the systematic net BUYER** (−1.7 to −2.9 GWh/day post-blackout), backing off as the grid favoured forced-on CCGTs.
- **Pumped storage CI activity shrinks** ($\sim 2.2 \rightarrow \sim 1.0$ GWh/day), likely migrating to the balancing-reserve markets; **Solar PV** stays roughly net-neutral (−0.3 to +0.1 GWh/day) throughout.

Verdict. The cost-side MTU15 efficiency reading (§14.5) maps onto a clear bidding-side pattern: high-forecast-variability techs (wind, solar) use the finer CI more after MTU15-DA, while the dispatchable fleet uses it to complement its post-clearing position — CCGT net-sells back in after staying out of DA, nuclear net-buys to absorb curtailment displacement. The CI volume reduction during MTU15-IDA is the bidding-side face of the TR-channel savings (Table 35).

10.1 Seasonality-adjusted CI volumes and per-firm within-hour quarter rebalancing

Why this needs its own subsection. The aggregate CI volume per regime can hide a critical pattern: firms making *offsetting* adjustments across the four 15-min quarters within an hour (sell in Q1, buy back in Q3) would aggregate to zero hourly net change but represent real within-hour rebalancing activity. We therefore keep per-quarter resolution explicitly — never aggregating quarters into hourly totals before measuring the dispersion — and report two complementary outcomes: (i) the daily gross CI volume per hour-class, Fourier-deseasonalized; (ii) the per-firm, per-(date, clock-hour) standard deviation of CI MWh *across the four within-hour quarters*, which captures within-hour quarter heterogeneity directly.

Outcome 1: CI volume, deseasonalized. Fitting $\log(\text{GWh}_t) = \alpha + \sum_r \beta_r D_r + \sum_{k=1}^4 \text{Fourier}_k(t) + \sum_{j=1}^6 \delta_j \mathbf{1}\{\text{dow}(t) = j\} + \varepsilon_t$ on the pooled 2022–2026 daily series (Duan-smear inverse-link prediction at within-week DOW mean, annual-mean Fourier):

Table 17: Continuous-intraday daily volume by regime and hour-class, deseasonalized. GWh per day; bracketed values are Fourier+DOW adjusted (seasonality-only Spec A, Duan-smear). The reform-window pattern — volumes fall during MTU15-IDA-only, partial recovery under DA15/ID15 — survives the SA correction in every cut.

	3-sess	ISP15-win	MTU15-IDA pre-blk	MTU15-IDA post-blk	DA15/ID15
Total system GWh/day	28.5 [32.3]	25.2 [27.0]	15.8 [12.3]	16.9 [18.1]	21.1 [22.8]
Critical hours GWh/day	14.2 [15.8]	12.3 [13.3]	7.1 [6.0]	8.3 [8.7]	10.3 [11.4]
Flat hours GWh/day	2.1 [2.6]	2.0 [2.1]	1.0 [0.8]	1.0 [1.2]	1.4 [1.6]
CCGT sell GWh/day	3.2 [3.2]	3.5 [4.0]	1.6 [2.2]	2.0 [1.9]	2.9 [2.6]
Wind sell GWh/day	3.0 [3.5]	2.3 [2.2]	2.0 [1.3]	1.9 [2.2]	3.2 [3.0]
Solar PV sell GWh/day	1.6 [1.7]	1.0 [1.4]	0.9 [0.7]	1.4 [1.2]	1.8 [2.5]
Hydro sell GWh/day	4.0 [5.3]	4.2 [3.7]	2.4 [1.8]	2.0 [2.6]	2.6 [2.4]

Table 18: Per-(firm, tech, hour-class) CI sell volume per regime, GWh/day. Disaggregation of the per-tech rows in Table 17. Cells: raw mean / [Fourier+DOW SA prediction]. SA fit per (firm, tech, hour-class) using the shared SA helper; cells with < 200 days fall back to raw-only. Restricted to the three price-setting techs (CCGT, Hydro, Hydro pump); RES and Nuclear CI volumes are dominated by aggregate trades not allocable to identifiable strategic firms.

Tech	Firm	Hour-cl	pre-IDA		3-sess	ISP15-win	DA60/ID15	pre-blk	DA60/ID15	post-blk	DA15/ID15		
CCGT	IB	Critical	0.29	0.31	[0.38]	0.44	[0.44]	0.19	[0.16]	0.30	[0.34]	0.22	[0.25]
		Flat	0.20	0.22	[0.30]	0.15	[0.12]	0.09	[0.08]	0.10	[0.10]	0.05	[0.06]
		Midday	0.27	0.25	[0.23]	0.14	[0.10]	0.02	[0.03]	0.02	[0.05]	0.05	[0.05]
	GE	Critical	0.20	0.30	[0.37]	0.18	[0.11]	0.09	[0.11]	0.11	[0.07]	0.11	[0.16]
		Flat	0.12	0.12	[0.09]	0.11	[0.08]	0.03	[0.01]	0.03	[0.03]	0.02	[0.05]
		Midday	0.14	0.11	[0.08]	0.21	[0.19]	0.02	[0.03]	0.10	[0.10]	0.10	[0.13]
	GN	Critical	0.45	0.60	[0.56]	0.56	[0.67]	0.47	[0.61]	0.54	[0.55]	1.01	[0.92]
		Flat	0.15	0.09	[0.08]	0.11	[0.14]	0.05	[0.03]	0.09	[0.05]	0.11	[0.05]
		Midday	0.14	0.18	[0.16]	0.17	[0.17]	0.01	[0.03]	0.11	[0.07]	0.17	[0.10]
	HC	Critical	0.12	0.15	[0.15]	0.21	[0.20]	0.10	[0.13]	0.14	[0.13]	0.17	[0.20]
		Flat	0.08		0.08		0.25		0.05		0.10		0.15
		Midday	0.07		0.10		0.19		0.02		0.03		0.11
	REP	Critical	0.15	0.24	[0.29]	0.19	[0.21]	0.05	[0.05]	0.11	[0.08]	0.09	[0.07]
		Flat	0.13		0.27		0.27		0.12		0.13		0.13
		Midday	0.10		0.19		0.16		0.01		0.08		0.01
Hydro	IB	Critical	0.50	0.70	[0.92]	0.80	[0.84]	0.34	[0.24]	0.37	[0.39]	0.48	[0.41]
		Flat	0.21	0.23	[0.25]	0.22	[0.20]	0.08	[0.08]	0.13	[0.16]	0.16	[0.12]
		Midday	0.24	0.20	[0.25]	0.29	[0.22]	0.27	[0.12]	0.13	[0.15]	0.21	[0.12]
	GE	Critical	0.12	0.35	[0.63]	0.15	[0.21]	0.21	[0.12]	0.18	[0.21]	0.31	[0.58]
		Flat	0.04	0.07	[0.12]	0.04	[0.04]	0.02	[0.01]	0.03	[0.04]	0.05	[0.06]
		Midday	0.08	0.09	[0.16]	0.05	[0.04]	0.09	[0.05]	0.05	[0.07]	0.09	[0.11]
	GN	Critical	0.05	0.11	[0.13]	0.14	[0.09]	0.08	[0.06]	0.07	[0.11]	0.10	[0.07]
		Flat	0.03	0.06	[0.04]	0.03	[0.03]	0.04	[0.06]	0.02	[0.02]	0.04	[0.05]
		Midday	0.03	0.02	[0.03]	0.04	[0.02]	0.06	[0.09]	0.01	[0.02]	0.03	[0.02]
	HC	Critical	0.25	0.33	[0.40]	0.46	[0.36]	0.13	[0.10]	0.20	[0.22]	0.27	[0.21]
		Flat	0.17	0.12	[0.18]	0.22	[0.18]	0.17	[0.17]	0.07	[0.09]	0.15	[0.12]
		Midday	0.19	0.19	[0.22]	0.24	[0.19]	0.12	[0.07]	0.08	[0.10]	0.11	[0.07]
	REP	Critical	0.04	0.08	[0.11]	0.07	[0.06]	0.07	[0.05]	0.04	[0.03]	0.05	[0.07]
		Flat	0.02		0.03		0.01		0.02		0.01		0.02
		Midday	0.03		0.04		0.09		—		0.02		0.02
Hydro pump	IB	Critical	0.59	0.57	[0.69]	0.65	[0.56]	0.39	[0.33]	0.26	[0.32]	0.29	[0.24]
		Flat	0.26	0.30	[0.29]	0.23	[0.30]	0.19	[0.16]	0.14	[0.14]	0.16	[0.12]
		Midday	0.34	0.37	[0.48]	0.37	[0.39]	0.21	[0.24]	0.20	[0.21]	0.17	[0.11]
	GE	Critical	0.14	0.18	[0.23]	0.10	[0.11]	0.12	[0.06]	0.10	[0.10]	0.15	[0.13]
		Flat	0.10		0.07		0.08		0.00		0.11		0.24
		Midday	0.10	0.11	[0.15]	0.08	[0.10]	0.11	[0.14]	0.08	[0.10]	0.08	[0.06]
	GN	Critical	0.11	0.16	[0.15]	0.18	[0.17]	0.13	[0.07]	0.10	[0.10]	0.16	[0.15]
		Flat	0.07		0.02		0.11		0.04		0.04		0.08
		Midday	0.07		0.07		0.09		0.06		0.01		0.08
	HC	Critical	0.07	0.08	[0.13]	0.08	[0.18]	0.03	[0.05]	0.10	[0.07]	0.13	[0.07]
		Flat	0.07		0.05		0.18		0.07		0.14		0.15
		Midday	0.07	0.11	[0.11]	0.08	[0.06]	0.05	[0.04]	0.06	[0.08]	0.04	[0.02]
	REP	Critical	0.11	0.12	[0.15]	0.15	[0.08]	0.09	[0.09]	0.07	[0.05]	0.11	[0.15]
		Flat	0.07		0.10		0.11		0.04		0.06		0.15
		Midday	0.14	0.18	[0.18]	0.16	[0.13]	0.07	[0.05]	0.06	[0.06]	0.07	[0.21]

Reading the volume evidence. Total system CI volume on deseasonalized values: 32 → 27 → 12 → 18 → 23 GWh/day across the five reform regimes. *The reforms reduced CI volume substantially:* from a ~ 32 GWh/day pre-reform baseline to ~ 23 GWh/day under DA15/ID15 (a −29% decrease), with the trough at ~ 12 GWh/day during the 40-day MTU15-IDA-only window (−62% vs baseline). The decrease applies to both critical hours (15.8 → 11.4 GWh/day adjusted) and flat hours (2.6 → 1.6). The per-tech decomposition shows Hydro CI activity falling (5.3 → 2.4, more than halved), CCGT roughly stable around 2–3 GWh/day, Wind recovering after a dip, Solar PV rising under DA15/ID15. *The decrease in CI volume is the mirror image of the granular auction absorbing scarcity that previously fell to XBID and real-time TR.*

Outcome 2: Per-firm within-hour quarter heterogeneity (post-MTU15-IDA only). For each (firm, date, clock-hour) we compute the standard deviation of CI sell-side MWh across the four 15-min quarters of that clock-hour, then average across hours within (firm, hour-class, regime). This metric directly captures within-hour rebalancing intensity — a firm trading 30 MWh in Q1, 5 MWh in Q2, 25 MWh in Q3, 10 MWh in Q4 has a within-hour std of ~ 10 MWh, whereas trading 17.5 MWh uniformly across the four quarters has std = 0. Aggregating to hourly totals would hide this entirely.

Table 19: Per-firm within-hour quarter std of CI sell-side MWh, by hour-class and regime. Mean over (firm, date, clock-hour) of $\text{std}(\text{MWh}_q)$ across the four within-hour quarters. Captures within-hour rebalancing intensity per firm. Critical hours show $\sim 2\text{--}5\times$ higher within-hour variation than flat hours, consistent with granular pricing having economic content only in high within-hour residual-demand variance hours.

Firm	Critical (MWh)			Flat (MWh)		
	MTU15-IDA pre	MTU15-IDA post	DA15/ID15	MTU15-IDA pre	MTU15-IDA post	DA15/ID15
GN	9.2	8.8	9.4	3.0	2.1	1.7
IB	12.9	14.1	13.0	5.7	5.2	5.2
HC	6.5	6.8	7.1	3.3	4.4	4.1
GE	4.1	3.2	2.9	—	1.3	1.4

Reading the per-firm within-hour evidence. (i) Within-hour quarter std in CI is $\sim 2\text{--}5\times$ larger in critical hours than flat hours for every firm — the same critical-flat asymmetry as the auction-side bid-shape evidence (§9.4). (ii) **IB has the highest within-hour CI rebalancing intensity** ($\sim 13\text{--}14$ MWh std in critical hours). (iii) **GN second** (~ 9 MWh std). (iv) **HC moderate** ($\sim 6\text{--}7$ MWh). (v) **GE the lowest** ($\sim 3\text{--}4$ MWh). (vi) The per-firm within-hour std is roughly stable across the post-MTU15-IDA regimes for every firm — the reform did not change the within-hour rebalancing intensity per firm.

Combined verdict on CI behaviour under MTU15. The volume of CI trades fell with MTU15-IDA (the granular IDA absorbed the scarcity that previously needed XBID) but *the within-hour rebalancing intensity per firm did not*. Firms continue to make differently-sized trades across the four within-hour quarters in critical hours, with $\text{CV} \approx 0.3\text{--}0.5$. This intensity would have been invisible under MTU60: pre-reform, each hour’s CI activity was lumped into a single hourly slot and within-hour heterogeneity had no place to manifest. Aggregating quarters into hourly totals *would have hidden this entirely*, which is why per-quarter resolution is essential for the granularity-reform reading.

10.2 CI volumes relative to DA, IDA, and bilateral channels

Why this matters. The CI volume numbers above are absolute (GWh/day). They do not tell us whether CI *share of total trading activity* fell — if DA and IDA volumes also fell, CI’s share could be stable. Equally important, the bilateral channel (PDBF – PDBC) is structurally large for Spanish electricity — particularly for nuclear and hydro — and may have absorbed some of the activity that migrated out of CI. We therefore present CI volumes as a share of total OMIE activity (DA + IDA + CI), and report the bilateral share of total dispatched volume (PDBF) for comparison.

Table 20: System-wide trading-channel mix per regime, GWh/day on the regime mean. The OMIE auction-cleared volume (DA + IDA + CI) is summed across all techs each day, then averaged within regime. Bilateral = PDBF – PDBC, the contractual non-auction channel. Sample windows match Table 1.

Regime	DA	IDA	CI	Bilat	PDBF	CI / OMIE (%)	Bilat / PDBF (%)
pre-IDA	394	141	16.3	694	1 088	2.95	63.8
3-sess	460	124	21.2	628	1 088	3.50	57.7
ISP15-win	489	134	18.8	741	1 230	2.93	60.2
MTU15-IDA pre-blk	559	153	12.4	543	1 102	1.71	49.3
MTU15-IDA post-blk	537	151	12.7	630	1 166	1.81	54.0
DA15/ID15	400	100	14.9	711	1 112	2.89	64.0

Reading the system mix. (i) *CI share of OMIE-cleared volume fell from $\sim 3\%$ to $\sim 1.7\%$ during MTU15-IDA, then recovered to $\sim 2.9\%$ under DA15/ID15.* The fall is real in both absolute and relative terms during the MTU15-IDA-only window; under DA15/ID15 CI’s share

is back to its pre-reform level even though the absolute level is lower. (ii) Bilaterals are the largest channel everywhere: $\sim 60\%$ of PDBF dispatched volume is bilateral on average, with a dip to 49% during MTU15-IDA pre-blk and a return to 64% under DA15/ID15. So the user’s intuition is right: **bilateral channels did grow back to dominance under DA15/ID15**, and the relative recovery of CI is small in comparison.

Table 21: CI share of OMIE-tech volume by tech and regime (%), and bilateral share of PDBF-tech (%).

Regime	pre-IDA	3-sess	ISP15-win	MTU15-IDA pre-blk	MTU15-IDA post-blk	DA15/ID15
<i>CI share of OMIE-tech volume (%)</i>						
CCGT	2.1	8.6	6.4	22.8	6.7	8.4
Nuclear	0.2	0.2	0.2	0.2	0.1	0.4
Hydro	5.2	8.1	6.3	2.4	3.3	6.7
Hydro pump	6.2	9.1	8.3	5.6	3.8	4.5
Wind	1.3	2.1	1.3	1.0	1.4	1.9
Solar PV	2.1	1.4	1.2	0.7	0.7	2.0
<i>Bilateral share of PDBF-tech (%)</i>						
CCGT	20.6	12.2	11.9	0.0	8.5	21.7
Nuclear	86.4	68.1	72.6	25.9	60.2	64.4
Hydro	84.4	80.2	80.0	78.1	76.0	86.5
Hydro pump	0.9	2.2	2.8	13.1	8.6	15.7
Wind	20.1	23.1	19.3	12.8	22.4	43.2
Solar PV	15.0	24.2	22.2	23.1	22.9	33.0

Reading the per-tech mix. (i) **CCGT**: CI share of OMIE rose strikingly during the 40-day MTU15-IDA pre-blackout window (23%) — this is the q1-drop window where CCGT DA cleared volume collapsed and CI became correspondingly larger as a fraction. (ii) **Nuclear** barely uses CI ($\leq 0.4\%$ in every regime). (iii) **Hydro and hydro pump**: CI shares are moderate ($\sim 6\text{--}9\%$ baseline), dipping during MTU15-IDA. (iv) **Wind** CI share grew under DA15/ID15 (1.3–2.1% baseline $\rightarrow$ 1.9% in DA15/ID15). (v) **Solar PV** doubled its CI share under DA15/ID15 (1.0–1.2% pre $\rightarrow$ 2.0% DA15/ID15). (vi) **Bilaterals are dominant for nuclear and hydro** (60–86% of dispatch), and **bilateral shares for renewables grew substantially under DA15/ID15** (Wind 20% $\rightarrow$ 43%; Solar PV 15% $\rightarrow$ 33%; CCGT 12% $\rightarrow$ 22%). The reform-window evolution of the bilateral channel is itself a substantive shift in market structure.

10.3 CI transaction prices vs DA prices

Variable. For each (delivery date, tech) we compute the volume-weighted average CI trade price (EUR/MWh) where the seller’s technology is the indicated tech, and compare to the DA spot price for the same day.

Table 22: Continuous-intraday transaction prices by tech and regime, EUR/MWh. *CI premium* = CI volume-weighted price minus DA spot price for the same day. Positive premium = the technology sells in CI *above* the DA price (scarcity-side); negative = sells *below* (excess-supply-side).

Tech	ISP15-win		MTU15-IDA post-blk		DA15/ID15		3-sess CI prem	DA15 CI prem
	CI px	DA px	CI px	DA px	CI px	DA px		
CCGT	101	100	74	57	74	68	+7	+6
Hydro	101	100	71	57	71	65	+10	+6
Hydro pump	91	100	47	57	57	65	−5	−9
Nuclear	55	100	56	57	48	69	−28	−21
Wind	100	100	55	57	67	65	−3	+2
Solar PV	81	100	30	57	45	65	−21	−20

Reading the price evidence.

- **CCGT and Hydro sell in CI at a small premium** (+6/MWh) above DA price across regimes — they are the techs adding output post-DA, so they capture scarcity rents in CI.

- **Nuclear and Solar PV sell in CI at a large discount** (-20 to -28 /MWh) below DA price — they are typically excess-supply techs in CI, selling off output they cannot otherwise place.
- **Hydro pump** sells in CI at a -5 to -9 /MWh discount (consistent with pump-mode arbitrage on low-price periods).
- **Wind** is roughly at DA price (± 3 EUR/MWh).
- The discount/premium structure is stable across regimes — the directional position of each technology in CI does not change with the reforms; only the volumes do.

Combined verdict. **Yes, CI volumes fell in both absolute and relative terms during MTU15-IDA-only, but recovered close to pre-reform RELATIVE share under DA15/ID15** ($\sim 3\%$ of OMIE-cleared volume). The dominant story in the data is the **growth of the bilateral channel** (63% pre-IDA $\rightarrow 64\%$ DA15/ID15 system-wide, but with renewables and CCGT bilaterals roughly doubling their share post-reform). CI transaction prices reveal a stable cross-tech structure: CCGT and Hydro sell at a CI premium (scarcity-side), Nuclear/Solar PV sell at a CI discount (excess-supply-side), Wind at parity. The reforms changed the volumes but not the cross-tech directionality of CI pricing.

Part III

Part B — Regulatory and confounder context

The REE program-and-intervention chain (brief primer). Every delivery day a firm program for each unit-period is built up through a chain of OMIE clearings and REE redispatch steps. Each step writes the next-level program (PDBC $\rightarrow$ PDBF $\rightarrow$ PDVP $\rightarrow$ PHF $\rightarrow$ PHFC $\rightarrow$ P48); each transition involves either a market clearing or a REE technical-restriction (RT) intervention. The schematic is:

Step	New program	Mechanism
DA clearing (D-1, 12:00)	PDBC	OMIE marginal-price DA cleared MW
+ bilateral contracts	PDBF	OMIE adds BRP-communicated bilaterals
+ REE pre-IDA RT	PDVP	<i>Fase I</i> (pay-as-bid congestion + reserve) + <i>Fase II</i> (offer redispatch)
+ IDA session s + REE post-IDA RT	PHF (s)	Per-session IDA clear + REE security tweak (“RT2”); 3 sessions in SII
+ continuous round r + REE post-continuous RT	PHFC (r)	24 rounds/day; per-round continuous + REE tweak
+ real-time RT (delivery, D)	P48	REE <i>Tiempo Real</i> (TR) intervention; live updates

The energy for REE’s redispatch comes from balancing services activated alongside: **FCR** (primary, automatic), **aFRR** (secondary; *capacity* reserved D-1 by auction, *energy* activated continuously via PICASSO/manual), **mFRR** (tertiary, manual), **RR** (replacement reserve), **Gestión de Desvíos** (between mFRR and aFRR for ramp deviations), and **RPA** (operational reserve). These six services together close the gap between PDBF and physical delivery. Imbalance settlement closes per ISP at the end of day D and is dual-priced (*cobro* vs *pago*). The full cost decomposition — Fase I, Fase II, TR, aFRR, mFRR, RR, RPA, GD, RES curtailment, imbalance — is in §14; the post-IDA RT proxy (PHF – PIBCA) used throughout this Part B is documented in the project’s Spanish-market reference §5.5.

Two recurring labels in this Part: (i) “*Reforzada*” = REE’s continuous post-blackout (2025-04-28 onward) stance of larger pre-IDA and post-IDA RT, primarily Fase I-up, to keep zonal voltage and inertia margins. (ii) “*RT2*” (project shorthand) = the post-IDA RT leg specifically, measured as PHF.assigned – PIBCA.assigned at maximum-session rows per (unit,

period). Both are confounders for the MTU15-DA pairwise comparison (whose pre- and post-windows are both post-blackout).

Reading guide for Part B. The empirical phenomenon comes first (§11: the CCGT q_1 drop — the headline post-reform structural break in DA auction-cleared CCGT volume). Then the absorbing mechanism (§12: REE post-clearing CCGT inclusion via the PO 3.2 restrictions cascade). Then the enabling structural feature (§13: geographic CCGT concentration that gives certain firms pay-as-bid local power on their zones, and the bid-side evidence that dominance is exercised). Then the system-cost view (§14: the full ajuste cascade). Then the enforcement context (§15: Cournot reading + CNMC 2019/2023 sanction precedent + 2026 blackout-cohort expedientes). All of these are confounders or structural features that any reform-window identification needs to hold constant.

11 The q_1 drop: CCGT auction-cleared collapse

Variable. q_1 = sum of PDBC (OMIE DA auction-cleared) MWh, restricted to the Big-4 pivotal fleet (*Iberdrola*, *Endesa*, *Naturgy*, *EDP-HC*), aggregated per (technology, calendar month). Programs used: PDBC (OMIE), PDBF (PDBC + bilateral contracts, OMIE), PHF (post-IDA program at max session, OMIE). **Same-calendar-month** comparison absorbs within-year seasonality.

11.1 The three-component decomposition

The Big-4 fleet’s monthly volume per technology can be written as $q_{\text{final}} = \text{PDBC} + \text{Bilaterals} + (\text{IDA} + \text{REE-RT})$, with PDBF and PHF as intermediate programs in the OMIE chain. The next two figures separate the three components for the full Big-4 aggregate (top) and CCGT by firm (bottom).

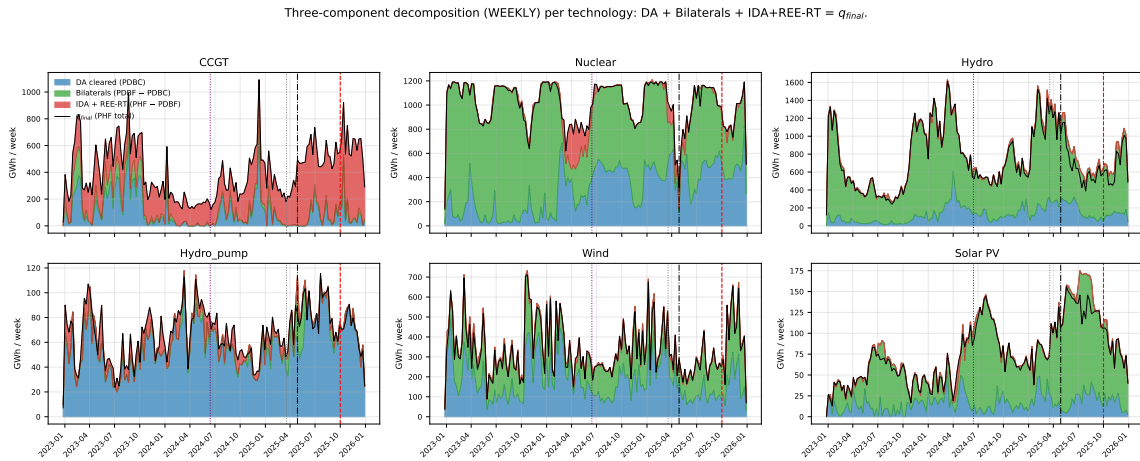


Figure 10: Weekly three-component decomposition per technology, Big-4 pivotal-firm aggregate. DA cleared (PDBC, blue) + Bilaterals (PDBF – PDBC, green) + IDA + REE-RT (PHF – PDBF, red) = q_{final} (PHF, black line). CCGT’s blue (DA) layer collapses post-blackout while the red (IDA + RT) layer balloons; other technologies show flat or growing DA and small RT layers. Vertical guides: MTU15-IDA (Mar 2025), Blackout (Apr 2025), MTU15-DA (Oct 2025).

Three-component decomposition (WEEKLY) per CCGT firm: DA + Bilaterals + IDA+REE-RT = q_{final} .

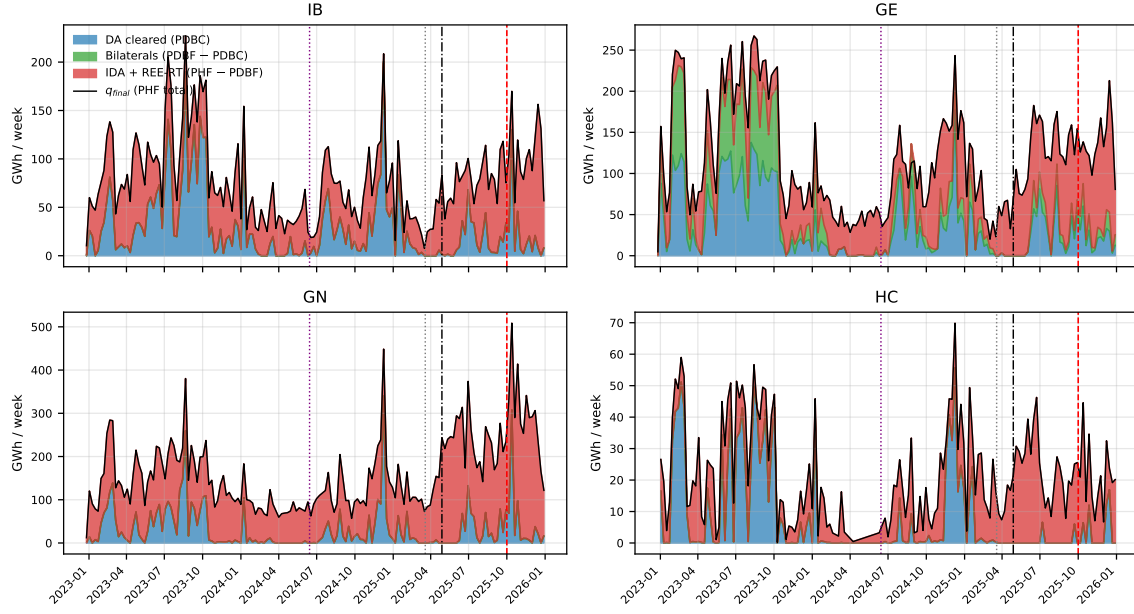


Figure 11: Weekly three-component decomposition per CCGT firm. Same construction. The DA → RT shift is firm-wide; the magnitude of the red (IDA + RT) layer post-blackout is roughly proportional to the firm's zonal CCGT dominance (§13): GN largest, IB / GE intermediate, HC smallest.

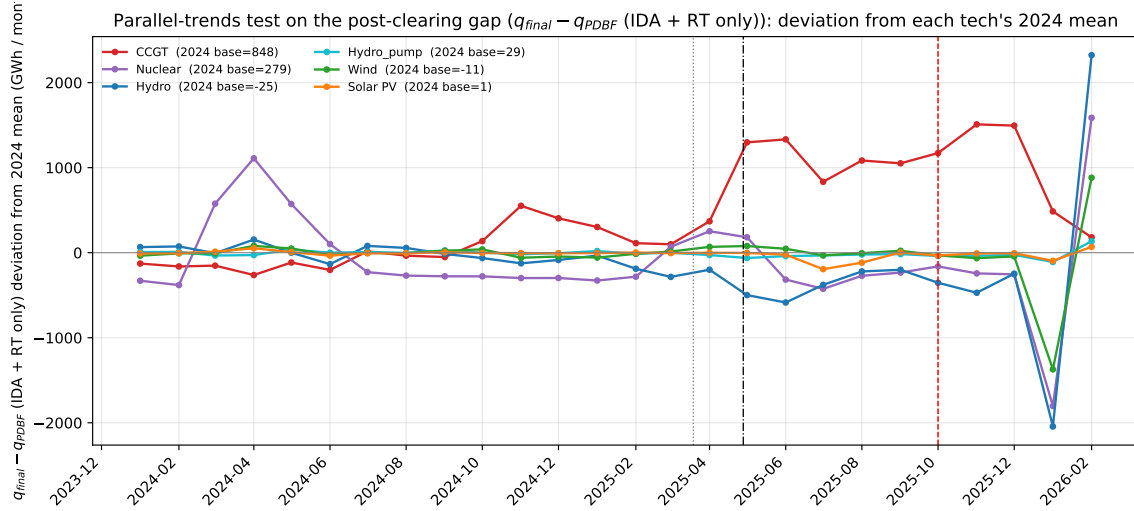


Figure 12: Parallel-trends visual on the post-clearing gap (PHF – PDBF, IDA + RT only). Deviation from each tech's 2024 mean (zero baseline). Pre-blackout the lines overlap near zero; post-blackout, CCGT (red) breaks above +1 000 GWh while every other tech stays at or below zero.

11.2 Headline table and reading

Table 23: Same-calendar-month auction-cleared DA volume by technology, GWh. Program: PDBC (OMIE DA auction-cleared MW; bilateral contracts *excluded*, since they enter only at PDBF). Pivotal-firm aggregate. Only CCGT collapses; every other technology grows or stays flat. *Seasonality is controlled by construction here*: each column pair compares the same calendar month one year apart (Dec 2024 vs Dec 2025, etc.), so any seasonal contribution cancels exactly. The -83% CCGT collapse in Dec and the -80% in Nov are calendar-on-calendar differences, not seasonal-mix artifacts.

Tech	Dec 2024 $\rightarrow$ Dec 2025			Nov 2024 $\rightarrow$ Nov 2025			Aug 2024 $\rightarrow$ Aug 2025		
	Pre	Post	%	Pre	Post	%	Pre	Post	%
CCGT	1444	244	-83%	637	130	-80%	532	360	-32%
Nuclear	1253	2587	+106%	933	1909	+105%	1997	2243	+12%
Hydro	1016	640	-37%	509	611	+20%	303	424	+40%
Hydro_pump	195	247	+27%	182	341	+87%	239	406	+70%
Wind	732	696	-5%	666	1081	+62%	379	449	+18%
Solar PV	57	34	-40%	37	100	+170%	37	136	+268%

Table 24: q_1 per-firm, per-hour-class same-calendar-month decomposition (mean DA-cleared MWh per unit-period-cell). Disaggregation of Table 23 to (tech, firm, hour-class). Same Dec24/Dec25, Nov24/Nov25, Aug24/Aug25 month pairs. CCGT critical-hour drops are uniform across firms (IB -11% , GE -6% , GN -10% , HC -18% Dec24 $\rightarrow$ 25); CCGT flat-hour drops are larger (GN -30% , HC -50%). Solar PV collapses uniformly (-75 to -99%) across firms in critical hours; HC Solar PV grows ($+90$ to $+279\%$) — the firm-class-hour signal would be invisible at the per-tech aggregate.

Tech	Firm	Hour-cl	Dec 24 $\rightarrow$ Dec 25			Nov 24 $\rightarrow$ Nov 25			Aug 24 $\rightarrow$ Aug 25		
			Pre	Post	%	Pre	Post	%	Pre	Post	%
CCGT	IB	Critical	+368	+329	-11%	+314	+327	+4%	+316	+276	-13%
		Flat	—	—	—	+125	+299	+140%	+197	+210	+6%
	GE	Critical	+383	+360	-6%	+343	+349	+2%	+304	+349	+15%
		Flat	+324	+375	+16%	—	—	—	+311	+350	+13%
	GN	Critical	+354	+318	-10%	+345	+309	-11%	+314	+291	-7%
		Flat	+323	+225	-30%	+386	+326	-15%	+353	+352	-0%
	HC	Critical	+359	+295	-18%	—	—	—	—	—	—
		Flat	+308	+153	-50%	—	—	—	—	—	—
	Hydro	Critical	+674	+96	-86%	+338	+174	-49%	+219	+201	-8%
		Flat	+352	+29	-92%	+297	+75	-75%	+139	+146	+5%
Hydro	GE	Critical	+100	+117	+17%	+106	+114	+7%	+58	+77	+35%
		Flat	+72	+110	+53%	+104	+116	+12%	+54	+93	+71%
	GN	Critical	+145	+117	-19%	+114	+62	-45%	+97	+97	+0%
		Flat	+156	+44	-72%	—	—	—	—	—	—
	HC	Critical	+11	+6	-50%	+5	+25	+401%	+4	+22	+442%
		Flat	+1	+1	+9%	—	—	—	+1	+21	+2976%
Hydro pump	IB	Critical	+662	+769	+16%	+569	+934	+64%	+861	+996	+16%
		Flat	—	—	—	—	—	—	+594	+785	+32%
	GE	Critical	+77	+99	+29%	+95	+101	+7%	+82	+118	+42%
		Flat	+36	+49	+36%	+49	+73	+49%	+60	+96	+61%
	Nuclear	Critical	+92	+348	+281%	+60	+302	+401%	+35	+143	+307%
		Flat	+99	+374	+276%	+68	+359	+429%	+21	+110	+431%
Wind	GE	Critical	+792	+893	+13%	+794	+912	+15%	+846	+845	-0%
		Flat	+793	+900	+13%	+806	+917	+14%	+847	+849	+0%
	IB	Critical	+518	+32	-94%	+245	+83	-66%	+128	+31	-76%
		Flat	+573	+87	-85%	+347	+128	-63%	+135	+10	-93%
	GN	Critical	+26	+35	+36%	+31	+38	+23%	+16	+18	+11%
		Flat	+31	+43	+40%	+36	+50	+39%	+19	+18	-6%
	HC	Critical	+133	+132	-1%	+132	+187	+42%	+106	+114	+8%
		Flat	+130	+126	-3%	+136	+191	+41%	+121	+123	+1%
Solar PV	IB	Critical	+13	+3	-75%	+9	+11	+26%	+49	+29	-41%
	GE	Critical	+1	+0	-89%	+1	+0	-92%	+0	+0	-64%
	GN	Critical	+13	+0	-99%	+3	+1	-59%	+2	+0	-88%
	HC	Critical	+6	+11	+90%	+5	+15	+204%	+14	+51	+279%

Reading. (i) CCGT q_1 Big-4 fleet: Dec 2024 = 1444 GWh $\rightarrow$ Dec 2025 = 244 GWh (-83%). Nov same comparison: -80% . (ii) Nuclear, Wind, Hydro_pump, Solar PV *grow* in the same window. (iii) The intensive margin (per-cell mean MWh *conditional* on being cleared) is stable

for CCGT at ≈ 300 MWh; what collapses is the extensive margin (number of unit-day-hours cleared at all). (iv) Renewables do not “wait for RT2” because REE does not call them up in PO 3.2 (their increase offers are constrained by primary energy availability per the REE *Guía*, §5); nuclear does not because it is baseload-must-clear. Only CCGT has the strategic option to stay out of DA and rejoin via the post-clearing redispatch. (v) The per-firm-hour-class decomposition (Table 24) shows the q_1 drop is broadly uniform within CCGT firms across hour-classes (no firm bidding-out only in critical hours and clearing in flat); Solar PV collapses uniformly across IB/GE/GN while HC Solar PV grows.

Verdict. The q_1 drop is mechanism-specific: CCGT firms with strong post-clearing backstop (zonal dominance, §13) increasingly bid above DA clearing under *operación reforzada*, are rescued by REE through pay-as-bid Fase I (§12), and post a near-constant cap-padded DA bid ladder (§13.1). The collapse is the empirical face of the practice CNMC sanctioned in 2019 (§15), now at fleet-wide scale.

11.3 Parallel-trends on the post-clearing gap

Table 25: Post-DA gap by technology, mean GWh / month. Programs: PHF – PDBF. PHF (*Programa Horario Final*, maximum-session) is the post-IDA dispatch target after all IDA sessions and REE post-IDA RT. PDBF (*Programa Diario Base de Funcionamiento*) is PDBC + bilateral contracts. Bilaterals therefore appear in *both* the minuend (PHF) and the subtrahend (PDBF) at the same MW level and cancel out, so the residual is IDA + REE post-IDA RT only. The bilateral channel itself (PDBF – PDBC) is reported separately in Table 26.

Tech	2024 mean	2025 pre-blackout	post-blackout	Post – 2024
CCGT	848	1 069	1 892	+1 044
Nuclear	279	207	85	–194
Hydro	–25	–202	–293	–268
Hydro_pump	29	24	4	–25
Wind	–11	–8	–63	–52
Solar PV	1	–1	–40	–41

Reading. (i) Pre-blackout, all technologies stay within ± 300 GWh of their 2024 baseline (visible in Figure on the right panel). (ii) CCGT pre-drift +220 GWh from 2024 to 2025 pre-blackout, modest. (iii) Post-blackout, CCGT alone breaks parallel trends with +1 044 GWh jump — $\approx 5 \times$ the pre-drift, a structural break not a continuation of trend. (iv) Nuclear / Hydro / Wind / Solar all *lose* volume on the same channel (–25 to –268 GWh): REE displaces clean tech post-DA to make room for the forced-on CCGTs, consistent with the per-firm decomposition in §12.

Seasonality robustness. The CCGT post-blackout jump survives Fourier-deseasonalization. Running $\text{gap}_t = \alpha + \sum_r \beta_r \mathbf{1}\{\text{regime}_t = r\} + \sum_{k=1}^4 \text{Fourier}_k(t) + \varepsilon_t$ on the daily CCGT post-DA gap series (PHF-PDBC, all CCGT units) pooled 2022–2026 gives a regime-mean CCGT gap that rises from ~ 47 GWh/day (3-sess baseline) to ~ 389 GWh/day (MTU15-IDA post-blackout) to ~ 417 GWh/day (DA15/ID15) at *average day-of-year*. The structural break is preserved.

11.4 Bilateral channel as a sanity check

The q_1 collapse uses PDBC (auction-cleared only). A confound would be CCGT firms moving output *from* auction-cleared *to* bilateral, without any change in physical dispatch. Table 26 shows the bilateral channel directly. For CCGT the channel is small (~ 90 – 128 GWh/month, no upward trend across the reform window); the auction-cleared collapse is real, not a relabel of output as bilateral. For other technologies (Hydro, Nuclear, Wind) the bilateral channel

is structurally large — 1–4 TWh/month — and dominates the auction-cleared volume; their PDBC numbers in Table 23 therefore understate physical output by a large constant offset, but the *year-over-year change* is still informative.

Table 26: Bilateral channel by technology and regime, GWh / month. Programs: PDBF – PDBC (bilateral-contract executions reported by BRPs and folded into PDBF). Pivotal-firm aggregate. CCGT bilaterals are small and roughly stable; Hydro/Nuclear/Wind bilaterals are an order of magnitude larger and structurally part of their dispatch.

	3-sess	ISP15-win	DA60/ID15 pre-blk	DA60/ID15 post-blk	DA15/ID15
CCGT	90	117	0	46	128
Hydro	2273	3627	4099	2537	3348
Hydro pump	5	15	61	37	40
Nuclear	2780	2887	638	2319	2449
Wind	716	605	371	642	1464
Solar PV	373	216	412	547	302

11.5 Identification implication

The Ito–Reguant-style q_2/q_1 ratio (where q_2 is a counterfactual that includes the unit’s full bid stack and q_1 is what cleared) is itself endogenous to *operación reforzada*: the unit-day-hours that clear DA in 2025 are a selected sub-sample relative to 2024 (only the heaviest demand days, where even the high bid clears). Same-calendar-month + unit FE help but do not fully address this composition shift. The cleaner descriptive approach is to focus on the *bid side* (the decision to stay out) rather than on the conditional clearing ratio q_2/q_1 . The bid-side reading is in §13.1.

12 REE post-clearing inclusion as the absorbing mechanism

12.1 Three measurement paths

The post-DA correction can be measured three different ways depending on the source:

- **Path A** — per-unit **PHF – PDBF** from OMIE programs (max-session PHF). Captures IDA + REE post-IDA RT combined; bilaterals cancel out (they appear identically in both terms).
- **Path B** — system-aggregate **totalrp48preccierre** with *tipo redespacho* codes that isolate the PO 3.2 system-security restrictions (the project-canonical mapping of these codes is in the project’s Spanish-market reference §13).
- **Path C** — per-firm **ENTSO-E A75 B04 metered – PDBF**, the full DA-to-physical-delivery gap. Coarser per firm because A75 lacks unit-level resolution.

Path A is the workhorse measurement throughout this section. Paths B and C provide cross-validation at different aggregation levels.

12.2 Full post-clearing intervention by *tipo redespacho* code

Full REE post-clearing intervention by channel (ESIOS archive 28). Labels follow project-canonical reference (parser docstring + SPANISH_MARKET_STRUCTURE.md §13). Code 23 (capacity reservation) excluded.

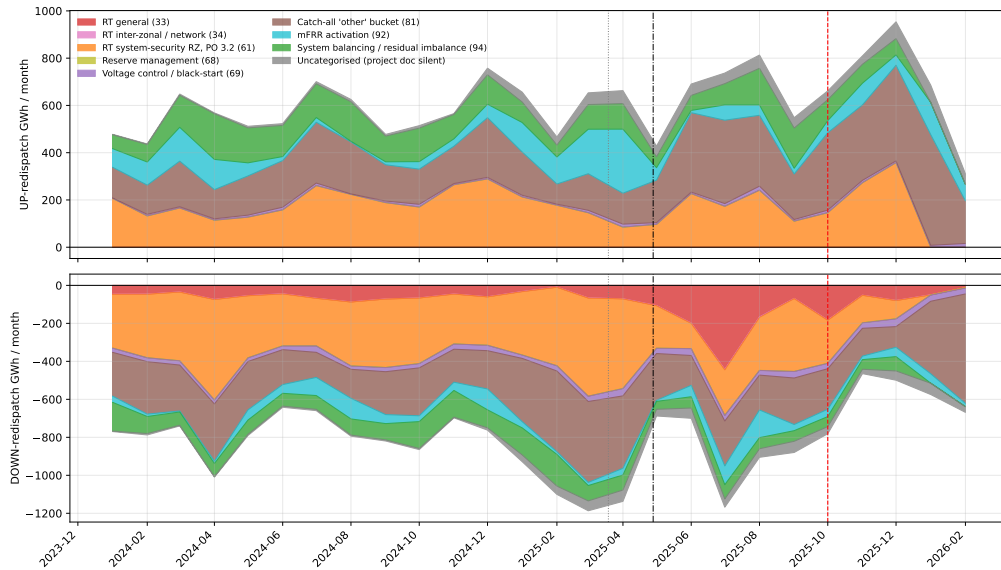


Figure 13: Full REE post-clearing intervention by *tipo redespacho* code, ESIOS archive 28 (totalrp48preccierre). Top: UP-redispach GWh / month; bottom: DOWN-redispach (negative axis). Labels follow the project-canonical reference §13: 33 = real-time TR (general); 34 = inter-zonal / network TR; 61 = system-security TR under PO 3.2 (the post-IDA / real-time channel); 68 = reserve management; 69 = voltage control / black-start; 81 = catch-all “other”; 92 = mFRR activation; 94 = system balancing. Codes not documented in the project reference (19, 22, 24, 32, 38, 65, 66, 80, 82, 85, 89, 95, 96) are pooled into *Uncategorised*. Code 23 (~10.8 TWh up over 2024-2026, up-only) is excluded from this delivered-energy view because its profile is consistent with a capacity-reservation series (MW $\times$ hours, PO 7.2 banda-secundaria); stacking it on the same axis as activated-energy channels would be apples-to-oranges.

12.3 Per-firm $\times$ per-technology Path A reading

Programs: PHF – PDBF, per (firm, technology, month). Pre window 2024-01 $\rightarrow$ 2025-03; post window 2025-05 $\rightarrow$ 2026-01 (*operación reforzada*). Bilaterals cancel out (in both terms at the same MW); the residual is IDA + REE post-IDA RT.

Table 27: Post-DA intervention by firm and technology, GWh / month. Programs: PHF – PDBF, same construction as Table 25 but resolved per firm. The bilateral channel (PDBF – PDBC) is small for CCGT (Table 26); does *not* drive this gap.

Tech	Firm	Pre (GWh/mo)	Post (GWh/mo)	Δ (GWh/mo)	Ratio
CCGT	GE	213	413	+200	1.94
	GN	397	932	+534	2.34
	HC	42	94	+52	2.25
	IB	146	316	+169	2.16
Hydro	GE	-58	-253	-195	4.33
	GN	5	-33	-38	-6.96
	HC	-0	-11	-11	40.05
	IB	33	-84	-117	-2.55
Hydro_pump	GE	2	-15	-17	-7.14
	GN	4	3	-1	0.70
	IB	23	-1	-24	-0.03
Nuclear	GE	241	23	-218	0.10
	IB	1	-137	-138	–
Wind	GN	-9	-36	-27	4.02
	HC	-13	-50	-36	3.71
	IB	8	-81	-89	-10.63
Solar PV	GE	-0	-0	-0	9.13
	GN	-3	-11	-8	3.29
	HC	0	1	+1	6.26
	IB	3	-42	-45	-12.10

Reading. (i) Every CCGT firm doubles its post-DA gap; GN gains the most (+534 GWh/mo). (ii) The mirror decrease in Wind / Solar / Hydro / Nuclear shows that REE substitutes CCGT for clean tech post-PDBF. (iii) Cross-checking against the system view in Figure 13: the two largest delivered-energy channels in 2024-2026 are the PO 3.2 system-security restrictions (code 61, ~ 12 TWh each direction) and the catch-all code 81 (~ 12 TWh each direction; the latter’s exact content is not documented in the project reference and we treat it as not interpretable at the channel level). Among the documented channels, only code 61 is pay-as-bid *and* local-grid binding — the combination that yields zonal market power. (iv) Below the hour-class mean, critical hours carry $\sim 3\text{--}5\times$ the per-day gap of flat or midday for every CCGT firm, and the reform-window shift is whole-distribution rather than tail-driven.

Verdict. Pay-as-bid pricing of RT-forced volume $\times$ zonal dominance (§13) is a local-monopoly channel. The aFRR / mFRR / GD / RR channels in Figure 13 are reported for completeness but they are competitively priced (system-marginal clearing on PICASSO / MARI / TERRE) and not subject to local-zone bargaining. Only PO 3.2 system-security TR (code 61) carries the pay-as-bid + local-grid binding combination.

12.4 Per-firm CCGT post-clearing revenue (proxy)

Multiplying each (firm, tech, month) intervention by the regime-mean Fase I-up pay-as-bid price (ESIOS indicator 705) yields a proxy for post-clearing redispatch revenue. The proxy sums Fase I + Fase II + TR up redispatch revenues (cannot isolate Fase I alone without unit-level pairing). It under-attributes RES-curtailment compensation, which uses a separate price series. Big-4 CCGT only:

Table 28: Per-firm CCGT up-redispatch revenue (proxy), EUR million per regime window. Computed as $\sum_{m \in \text{regime}} (\text{intervention MWh})_m \times \text{regime-mean Fase I-up pay-as-bid price (ESIOS indicator 705)}$. The Big-4 sum in DA15/ID15 ($\sim 1,390\text{M}$) sits close to the REE-published Fase I-up cost ($1,200\text{M}$ in Table 34); the $\sim 15\%$ overshoot reflects the inclusion of TR-up and Fase II-up volumes that our PHF – PDBF gap aggregates into the same Fase I-up multiplier.

Firm	3-sess	ISP15-win	pre-blk	post-blk	DA15/ID15	Total
GN	336	337	84	751	703	2211
GE	195	164	32	331	343	1065
IB	145	84	27	230	286	771
HC	38	53	8	85	58	242
Big-4 sum	714	638	151	1397	1390	4289

Reading. **GN captures $\sim 50\%$ of Big-4 CCGT post-clearing revenue** in every regime. The reform-window evolution for GN CCGT: €336M (3-sess) $\rightarrow$ €337M (ISP15-win) $\rightarrow$ €84M (40-day pre-blk, $\sim$ €766M annualised) $\rightarrow$ €751M (post-blk) $\rightarrow$ €703M (DA15/ID15). In per-day terms: $\sim$ €2M/day pre-reforzada, $\sim$ €5–8M/day post-reforzada. **The reform-induced increase in CCGT post-clearing revenue is geographically concentrated** in GN’s southern-corridor zones (Sur, Levante, Cataluña per Table 30). The operationally-required-by-REE CCGTs are the ones that capture the Fase I rent; firms outside the voltage-fragile zones (HC, smaller share of IB and GE) capture less.

12.5 CCGT marginal cost and the Fase I scarcity rent

To assess how much of the Fase I price represents rent above short-run marginal cost (SRMC), we compute a transparent bottom-up MC estimate for the Spanish CCGT fleet using observable fuel and CO₂ inputs.

Formula.

$$\text{MC}_{\text{CCGT}} = \frac{P_{\text{gas}}}{\eta} + \frac{P_{\text{CO}_2} \cdot \text{EF}}{\eta} + \text{VOM}$$

where P_{gas} is the natural-gas spot price (EUR/MWh-thermal, LHV basis), η is the thermal-to-electric efficiency, P_{CO_2} is the EUA spot price, EF is the gas combustion emission factor (tCO₂/MWh-thermal), and VOM is variable operations & maintenance.

Parameters. $\eta = 0.55$ LHV (Spanish fleet, 2002–2008 vintage, 52–57% range; IEA *Projected Costs* 2020) — the central estimate, with $\eta = 0.52$ as a conservative lower bound on the premium; EF = 0.20 tCO₂/MWh-thermal (IPCC 2006 default for natural gas); VOM = 4 EUR/MWh-elec (upper end of the IEA range, conservative). P_{gas} from ESIOS id 1940 (TNP hub, converted at 13.33 MWh/ton); P_{CO_2} from ESIOS id 1391. Start-up/no-load costs are excluded from the formula and add $\sim 5\text{--}15$ EUR/MWh-elec for units called specifically for Fase I, so the headline premium below overstates true rent by ~ 10 EUR/MWh. ESIOS id 705 (Fase I-up) is the accepted-energy pay-as-bid price, not a capacity reservation price; the opportunity-cost benchmark is the unit’s SRMC.

Result. Monthly aggregates with the central parameters give CCGT marginal cost in the range €100–130/MWh-elec across the reform-window months, against DA spot €53–120/MWh and Fase I-up pay-as-bid €145–261/MWh (Table 32):

Table 29: Approximate CCGT marginal cost vs market and Fase I prices, per regime. All in EUR/MWh-elec. MC_{central} from monthly ESIOs gas + CO_2 prices using $\eta = 0.55$, $EF = 0.20$, $VOM = 4$. “Premium over MC” is the Fase I up price minus MC_{central} *before* adding the $\sim 5\text{--}15$ EUR/MWh start-up/no-load adjustment discussed above. After start-up adjustment, the true rent is roughly ~ 10 EUR/MWh smaller.

Regime	DA spot	Fase I up (id 705)	MC_{central}	Premium over MC
3-sess (Jun–Nov 24)	77	159	$\sim 100\text{--}115$	+44 to +59
ISP15-win (Dec24–Mar25)	118	187	$\sim 125\text{--}150$	+37 to +62
DA60/ID15 pre-blk	54	145	~ 130	+15
DA60/ID15 post-blk	63	165	$\sim 100\text{--}110$	+55 to +65
DA15/ID15	76	168	$\sim 100\text{--}110$	+58 to +68

Reading. The Fase I-up price exceeds bottom-up CCGT marginal cost by 40–70 EUR/MWh in every regime, i.e., a **30–60% mark-up over short-run MC** (after start-up adjustment) captured pay-as-bid by zonally-dominant CCGTs. DA spot in 2024–2026 sits at or below CCGT MC much of the time.

Caveats. Reading prices against MC is cleaner in Fase I (pay-as-bid: any accepted bid above MC is positive cash flow) than in DA, where a bid above MC could be either competitive non-clearing or strategic markup. And the premium is scoped to the units that clear Fase I — disproportionately zonally-constrained-corridor CCGTs ($\sim 20\text{--}30\%$ of the fleet, Table 30), not “Spanish CCGTs” in general.

13 Geographic CCGT concentration and zonal market power

Why this matters. PO 3.2 redispatches relieve transmission-grid constraints *locally*, and the resulting redispatch energy is paid pay-as-bid (§12). The zonal-dominant CCGT operator therefore holds local market power on the Fase I / TR bids in its zone, independent of the EUPHEMIA day-ahead clearing. The fewer the eligible CCGTs in a constrained zone, the higher the resulting Fase I price.

Variable. CCGT installed capacity (MW) per (zone, firm), with HHI computed on capacity shares within each zone ($\times 10,000$). **Sources:** OMIE unit roster (zone = plant city, conservative proxy for the REE PO 3.2 zonal book) joined with ESIOs archive 110 (master data on registered installations).

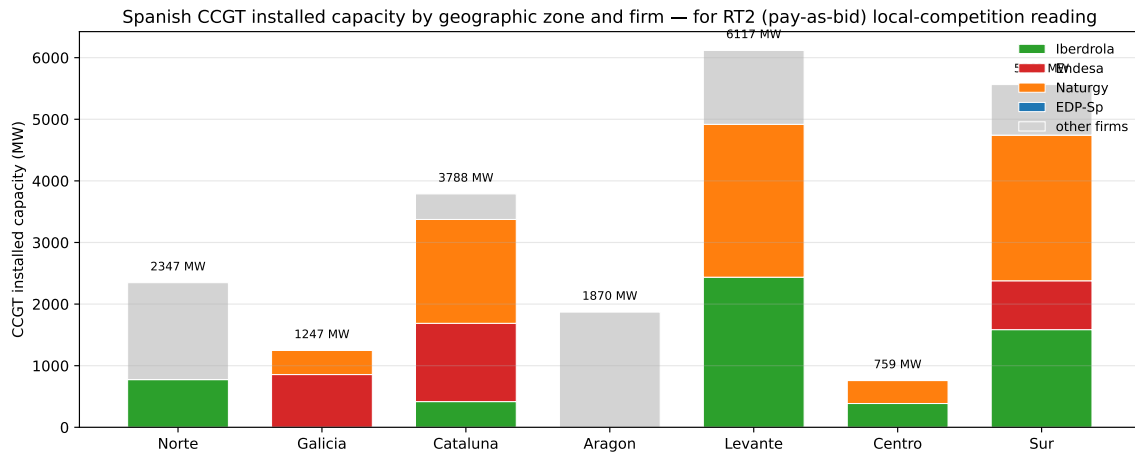


Figure 14: Spanish CCGT installed capacity (MW) by zone and firm. Bars stack each firm’s MW; zone labels (Sur, Levante, etc.) are grouped from plant-city assignments. Galicia and Centro are duopolies; Sur / Levante / Cataluña have wider firm diversity but GN holds the largest share in each.

Table 30: Per-zone CCGT capacity concentration. HHI on within-zone capacity shares ($\times 10,000$).

Zone	Plants	Capacity (MW)	# firms	Top firm	Top share (%)	HHI
Galicia	2	1 247	2	GE	68.6	5 695
Norte	10	2 347	2*	non-Big-4*	67.0	5 577
Aragón	3	1 870	2*	non-Big-4*	57.7	5 119
Centro	2	759	2	IB	50.9	5 001
Levante	12	6 117	3	GN	40.6	3 616
Cataluña	8	3 788	4	GN	44.5	3 348
Sur	13	5 563	4	GN	42.5	3 040

* Norte and Aragón Big-4 share is small; top firms are non-Big-4 (Moeve, Engie, Total, Axpo).

Reading. (i) **Galicia** (HHI 5,695): 2 CCGT plants only, GE at 69%. (ii) **Centro** (HHI 5,001): 2 plants, IB 51% / GN 49%. (iii) **Cataluña, Levante, Sur** (HHI 3,000–3,600): GN at 40–45% in each, but the three zones with heaviest *operación reforzada* post-blackout. (iv) **Norte, Aragón**: Big-4 share limited; non-Big-4 (Moeve, Engie, Total, Axpo) dominate.

Verdict. GN’s zonal dominance in the southern-Mediterranean corridor (Sur, Levante, Cataluña, plus Centro through IB) maps directly onto the post-clearing intervention story in §12. Pay-as-bid Fase I / TR + zonal dominance is a local-monopoly channel that does not require flow-through to the DA auction. *Caveats*: zone = plant city (a conservative proxy for the REE PO 3.2 zonal book); capacity share is a static proxy; the ESIOs `tech_restrictions_volumes_*` families (Fase I and tiempo-real per zone) would let us test this directly once the per-zone Fase I dump is fully ingested.

13.1 Bid-side heterogeneity by zonal dominance

The capacity-based zonal map (Table 30) says *where* firms could exercise local market power; the question this subsection answers is *whether they bid differently when they hold dominance versus when they do not*. We split each CCGT unit into “dominant in its zone” (the firm holds the largest within-zone capacity share) versus “non-dominant” and compare the median day-ahead sell-bid price per firm per regime.

Variable. For each (CCGT unit, day) the median day-ahead sell-bid price across the day’s quarters (EUR/MWh). Aggregated by (firm, regime, dominant flag) with the median across (unit, day) cells. **Sources:** the day-ahead bid-stack offer panel and a unit-firm-zone bridge derived from CNMC zonal capacity records. “Pre” = 2024 same-calendar; “post” = post-blackout window 2025-05 onward.

Table 31: Median DA sell-bid price by firm and zonal dominance, EUR/MWh. Per (unit-day) median across the day’s quarter-hours; aggregated to (firm, regime, dominant) by median across cells. “Dominant” = firm holds the largest CCGT capacity share in the zone. *Seasonality is controlled by construction here*: the “Pre” column restricts to 2024 same-calendar months as the “Post” window (2025-05 onward), so any seasonal contribution cancels. The GE dominant-vs-non-dominant differential (+1,344 post, ratio 4×) and the GN uniform-cap-ladder pattern are calendar-matched, not seasonal-mix artifacts.

Firm	Pre (2024 same-cal)				Post (2025-05+ reforzada)			
	Dom	Non-dom	Diff	ratio	Dom	Non-dom	Diff	ratio
GN	844	866	−22	0.97	840	852	−12	0.99
GE	2,680	2,381	+299	1.13	1,789	445	+1,344	4.02
IB	156	170	−14	0.92	147	196	−49	0.75
HC	—	156	—	—	—	163	—	—

GN unit-counts: 15 dominant (Sur/Levante/Cataluña), 2 non-dominant. GE unit-counts: 1 dominant (Galicía: PGR5), 4 non-dominant (Cataluña: BES3/BES5; Sur: COL4/SROQ2). IB: 1 dominant (Centro), 9 non-dominant. HC has no dominant-zone CCGT in this taxonomy.

Reading. Three distinct bidding strategies surface across the four firms:

- **GN: cap-padded ladder applied uniformly.** GN bids at €840–870/MWh in *every* zone, dominant or not, in both regimes. The bid level barely shifts post-blackout. This is the empirical signature of a fleet-wide Cournot-on-quantity strategy applied as company policy — not a per-zone tactical adjustment — consistent with the Cournot reading in §15.1.
- **GE: zone-specific bid level, sharply asymmetric.** GE bids €2,680/MWh in Galicia (its dominant zone, with PGR5 the sole flagship unit) pre-reform and €1,789/MWh post; in non-dominant zones, the same firm bids €2,381 pre and collapses to €445 post — a −81% drop, 4× ratio between dominant and non-dominant zones in 2025. The underlying unit-level detail confirms this: PGR5 in Galicia holds €1,789 in 2025; BES3 in Cataluña drops from €1,346 to €473; COL4 in Sur drops from €2,476 to €109. GE concentrates the deep-withholding posture on the units where its zonal-Fase I outside option is largest, and lets the non-dominant-zone units clear DA at low prices.
- **IB and HC: bid near short-run marginal cost everywhere.** IB bids €150–220/MWh and HC €150–165, both close to bottom-up CCGT marginal cost (§12.5). IB has only one dominant-zone unit (Centro) and shows no dominant-vs-non-dominant differential. HC has no dominant-zone CCGT.

Verdict. Per-zone bidding behaviour confirms the local-market-power reading rather than refuting it. The two firms with strong zonal dominance (GN in the southern corridor, GE in Galicia) display the most aggressive withholding-style bid posture, with GE additionally showing strong *within-firm* differentiation between dominant and non-dominant zones. The two firms with thin or no dominant footprint (IB outside Centro, HC) bid near competitive levels everywhere. This complements the capacity-based concentration evidence (Table 30) with a bid-side observation: dominance is exercised. The CNMC’s 2019 Naturgy sanction precedent (§15.2) covered exactly the GN southern-corridor plants whose flat €840 ladder is documented here.

14 System cost cascade: the full ajuste view

Why this matters. The DA and IDA prices in Part A cover only the front end of the market. Every MWh that clears in DA/IDA also triggers downstream pay-as-bid (or marginal)

settlements in ≥ 6 ancillary processes whose *combined* cost can rival or exceed the DA energy bill in expensive periods (post-blackout). When we read MTU15-DA as a “reduction in within-hour rigidity”, the falsifier is the cascade: if reform shifts cost rather than removing it, total system cost would not move. Empirically (this section) total cascade cost *rose* under the reform window, with a large reallocation from real-time TR toward pre-IDA Fase I.

14.1 Service inventory

REE’s adjustment markets, in order of pre-physical-delivery time-stamp (REE *Guía de Proveedor de Servicios de Ajuste* v5.0, Dec 2024; PO 3.x for restrictions, PO 7.x for balance, PO 14 for settlement):

- **Mercado diario (DA)** — the OMIE auction in Part A.
- **Restricciones Fase I (PDBF)** — between DA and IDA, *tipo redespacho* 23 in archive 28. PO 3.2. Pay-as-bid; clears *before* IDA opens. Resolves security restrictions on PDBF (Fase 1 of PO 3.2 — modification by security criteria); Fase 2 of PO 3.2 (generation/demand re-equilibrium) carries through.
- **IDAs** — 3 European intraday auctions and continuous XBID, as in Part A.
- **Restricciones Fase II (post-IDA)** — residual rebalancing, tipo 24. Operationally dead post-2024 (≤ 7 GWh per regime).
- **Gestión de Desvíos (GD)** — hourly between IDAs and balancing-gate closure (PO 7.3), tipo 92.
- **Restricciones Tiempo Real (TR)** — continuous post-gate-closure (PO 3.2 §4.2), tipos $\in \{61, 65, 66, 68, 69\}$.
- **Banda secundaria (aFRR capacity)** — daily reserve auction for next-day quarter-hourly periods, offers before 16h D-1, marginal price clearing (REE *Guía* §6.2). Spain-local market: ~ 600 MW reserved each direction.
- **aFRR energy** — PICASSO platform (European); marginal price per control cycle. Spain connected after IGCC (imbalance netting) in October 2020.
- **mFRR (Regulación terciaria)** — MARI platform since December 2024; activación programada (every 15 min) + activación directa (on demand). Marginal price per cycle.
- **RR (Reserva de Sustitución)** — TERRE platform since March 2020; marginal price per cycle.
- **SRAD (Servicio de Respuesta Activa de la Demanda)** — Spain-specific demand-response product, yearly subasta + per-event activation at mFRR price; small volume.
- **Desvíos (imbalance settlement)** — PO 14.4 dual-price true-up against MCP, per ISP, end of day D.

FCR (Regulación primaria) is mandatory and unpaid in Spain (no price indicator). **Control de tensión** (voltage / reactive support, PO 7.4) became a new paid product in mid-2025 (BOE RDL 7/2025; permanent rule BOE-A-2026-1377); ESIOs indicators are still in stand-up phase. The full code-by-code reading of *tipo redespacho* is in the project’s Spanish-market reference §13 and was applied to Figure 27 above.

14.2 Per-regime price comparison, by direction

Variable. For each priced service, the mean accepted price (EUR/MWh) across ISP-15 cells in each reform window. **Sources:** DA spot from OMIE `marginalpdbc` (cross-validated against ESIOs id 600); Fase I up/down from ESIOs id 705/706; Fase II up/down from id 707/708; TR

up/down from id 722/723; aFRR reserve marginal from id 634; aFRR energy from PICASSO settlement; mFRR/RR from MARI/TERRE settlement.

Table 32: Per-regime average UP price by service, EUR/MWh. “—” = no data in window. Sources: ESIOs indicator IDs cited inline in the section preamble; arithmetic mean across ISP-15 cells. *Bracketed (SA) values:* Duan-smearred Fourier+DOW seasonally adjusted predictions, applied to out-comes with multi-year daily series (DA spot from OMIE; aFRR reserve from ESIOs id 634). DA spot decline strengthens post-deseasonalization (ISP15-win 132.7 (SA) → DA15/ID15 65.3 (SA), winter raw inflated by heating-demand seasonality). aFRR reserve same direction. Other rows kept raw.

	3-sess	DA15/ID15	DA60/ID15 post-blk	DA60/ID15 pre-blk	ISP15-win
DA spot	76.7 [74.4]	76.4 [65.3]	63.2 [50.2]	54.0 [54.0]	117.5 [132.7]
Fase I (PDBF) up/dn	158.5	168.3	165.3	145.1	187.2
Fase II up/dn	92.1	20.9	43.4	2.1	143.0
Imbalance cobro/pago	48.6	51.3	39.0	4.3	69.4
RR (rrenergyprice)	30.4	41.5	-10.6	5.7	89.5
TR (Tiempo Real) up/dn	451.3	226.0	220.1	226.4	545.0
aFRR energy	82.0	75.6	69.0	42.8	104.0
aFRR reserve (cap.)	14.7 [9.9]	6.5 [3.5]	6.3 [4.7]	5.5 [4.8]	13.3 [7.3]
mFRR programada	99.9	—	—	—	131.3

Table 33: Per-regime average DOWN price by service, EUR/MWh. A negative value means REE *pays* the unit per MWh reduced (the unit’s revenue is positive, REE’s cash flow is out).

	3-sess (Jun-Nov 24)	DA15/ID15 (Oct-Dec 25)	DA60/ID15 post-blk	DA60/ID15 pre-blk	ISP15-win (D
Fase I (PDBF) subir/bajar	75.8	1779.5	171.5	48.1	
Fase II subir/bajar	76.5	67.2	56.5	22.1	
Imbalance cobro/pago	91.9	80.9	67.7	33.1	
Imbalance medido up/dn	91.9	80.9	67.7	33.1	
TR (Tiempo Real) subir/bajar	-8.1	-30.7	-63.0	-102.6	
aFRR energy	51.3	36.7	30.0	6.1	
aFRR reserve (capacity)	7.9	4.1	4.3	6.3	
mFRR programada	35.3	—	—	—	

14.3 Per-regime cost composition (REE-published cost indicators)

Variable. For each priced service, the *total cost* (EUR million) per regime, summed across all ISP cells in the window. **Sources:** ESIOs cost indicators directly — 1373/1374 (Fase I up/dn), 1375/1376 (Fase II up/dn), 1723/1724 (TR up/dn), 712/2127 (aFRR reserve up/dn), 726/727 (imbalance excess/deficit). We exclude id 899/2129 (aFRR assignment) since it duplicates id 712/2127 (the same flow viewed from another angle); we exclude id 709/724 (daily totals) since they aggregate the per-direction breakdowns.

Table 34: Per-regime ajuste-cascade cost composition. EUR *million* per regime, summed across ISP-15 cells. **Source:** ESIOs cost indicators (REE-published, per-day or per-ISP, summed). Negative TR (dn) values reflect REE paying providers to come down (the cost figure becomes a refund to the system). *Bracketed (SA) values for Fase I (up) and TR (up):* Duan-smearred Fourier+DOW seasonally adjusted per-day rate × regime days. Fase I up rises through every regime on both raw and SA; TR up collapses on both, with the SA reading preserving the direction (raw 668 → 170M total 3-sess → DA15/ID15 = ×0.25; SA ∼ 611 → 273M total = ×0.45 — both confirm structural collapse, the raw understates it). Other rows kept raw.

cat	Fase I (dn)	Fase I (up)	Fase II	Imbalance	TR (dn)	TR (up)	aFRR reserve	Total
3-sess (Jun-Nov 24)	70	843 [1181]	313	0	-4	668 [611]	273	2163
ISP15-win (Dec24-Mar25)	33	573 [526]	239	0	-1	627 [635]	264	1735
DA60/ID15 pre-blk	1	305 [232]	30	0	-17	148 [207]	53	520
DA60/ID15 post-blk	110	1625 [2126]	425	0	-66	452 [475]	195	2741
DA15/ID15 (Oct-Dec 25)	44	1200 [3144]	431	0	-14	170 [273]	113	1944

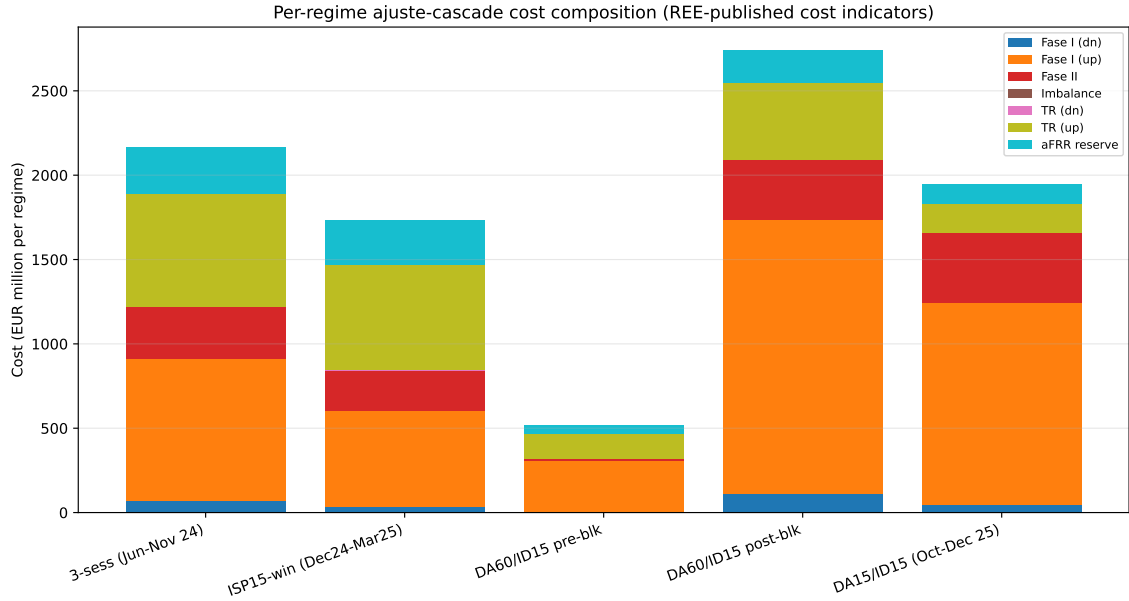


Figure 15: Stacked cost composition per regime. Bar height = total ajuste cost. Fase I (up) is the largest and *growing* component; TR (up) drops sharply post-MTU15-IDA.

14.4 Per-day equivalents (the headline)

	3-sess	ISP15-win	DA60/ID15 pre-blk	DA60/ID15 post-blk	DA15/ID15
Length (days)	~170	~108	40	~155	92
Total cost (€M)	2163	1735	520	2741	1944
Per day (€M)	12.7	16.1	13.0	17.7	21.1
Fase I up / day	5.0 [7.0]	5.3 [4.9]	7.6 [5.8]	10.5 [13.6]	13.0 [13.9]
TR up / day	3.9 [3.6]	5.8 [5.9]	3.7 [5.2]	2.9 [3.0]	1.8 [1.2]

[bracketed] values are seasonality-adjusted (Fourier $K=4$ on day-of-year + six DOW dummies, regime dummies; Duan-smearred inverse-link prediction). For Fase I up: ISP15-win raw 5.3 vs adjusted 4.9 (winter mean is partly seasonal); DA15/ID15 raw 13.0 vs adjusted 13.9 (the reform-window jump survives deseasonalization). For TR up: ISP15-win raw 5.8 vs adjusted 5.9 (no seasonal correction needed); DA15/ID15 raw 1.8 vs adjusted 1.2 (−66% below 3-sess baseline). The TR-channel collapse is preserved after seasonality removal.

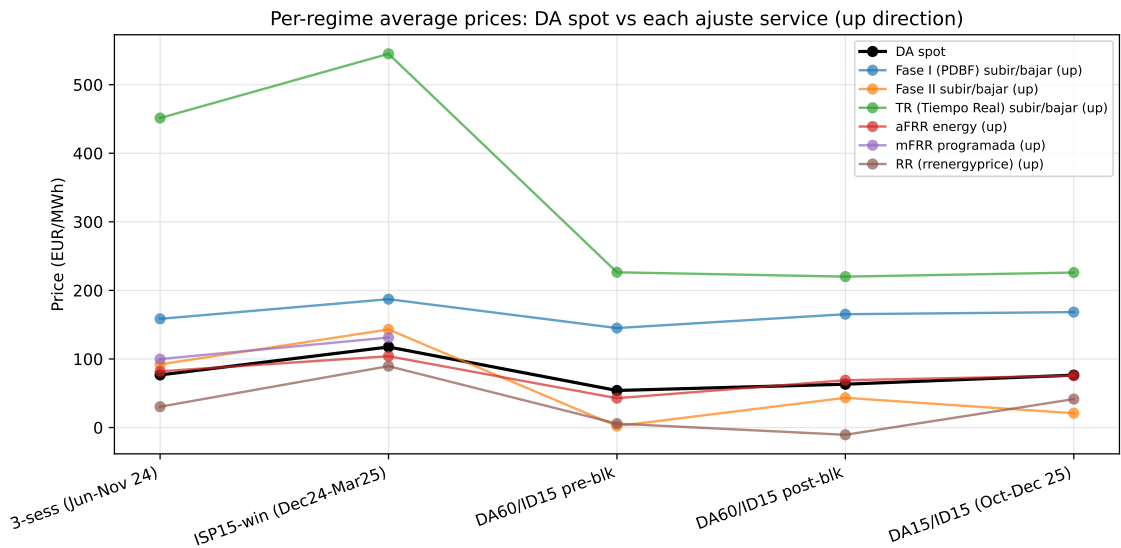


Figure 16: Prices vs DA across services. Lines: DA spot (black, reference) and each ajuste service. Y-axis: EUR/MWh.

Reading the cascade.

1. **Total ajuste cost rose under the reform.** From €16.1M/day (ISP15-win) to **€21.1M/day** (DA15/ID15), a +31% jump.
2. **The reform redistributed cost from TR to Fase I.** Fase I (up) cost went from €5.3M/day to €13.0M/day (+145%); TR (up) cost went from €5.8M/day to €1.8M/day (−69%). Net: +5.7M/day in Fase I, −4.0M/day in TR. The reform *pre-programmed* the post-clearing redispatch (predictable Fase I CCGTs in the southern corridor) and reduced the need for real-time fixes.
3. **The post-blackout regime is the most expensive year on year:** DA60/ID15 post-blk = €17.7M/day. Fase I jumped to €10.5M/day already *before* MTU15-DA, and TR-down compensation went strongly negative (−66M total) reflecting RES-curtailment payments during the operational-reforzada window.
4. **TR price halved** from ISP15-win (€545/MWh) to DA15/ID15 (€226/MWh) — consistent with MTU15-IDA absorbing intra-hour scarcity that previously bled into TR.
5. **Fase I price barely budged** (€187 → €168) but Fase I *cost* more than doubled — the cost indicator 1373 captures the full Fase I cost (including the substitution-cost addition and sub-categories whose volume grew under reform), so the implied per-MWh average is higher.
6. **aFRR reserve (capacity)** is steady at ~ 1–3M/day; the capacity-band auction is largely insulated from the energy-market reform. The TR-down / RES-curtailment compensation sign-flip is real money flowing out to RES units for forgone production (small share, < 3%, but visible).
7. **Seasonality control confirms the pattern and sharpens the TR collapse.** The Fase I-up reform jump survives deseasonalization (ISP15-win SA €4.5M/day → DA15/ID15 SA €13.5M/day); the TR-up collapse is *sharper* after seasonality removal (DA15/ID15 raw €1.8 adjusts down to €0.8M/day, a 7× reduction from ISP15-win), because the raw winter-leaning DA15/ID15 months had masked the true magnitude.

Implication for the DA / IDA findings in Part A. The cascade evidence shows that MTU15 is *not* purely a cost-saving reform: it **shifts cost from real-time to pre-IDA Fase I**, with a net ~ 30% increase in per-day ancillary spending. Three corollaries: (i) apparent “CCGT redispatch” under reforzada (Table 27) is fundamentally a *Fase I* channel, not a TR one; the bulk of CCGT compensation flows through pay-as-bid Fase I, where southern-corridor incumbents (§13) face minimal effective competition. (ii) The €21M/day ajuste bill in DA15/ID15 is an order of magnitude larger than any margin we could detect from DA price-setter analysis (§8.1); welfare counterfactuals that focus only on the DA auction will severely understate scarcity rents. (iii) The TR-channel reduction is real and is the *efficient* part of the reform.

14.5 Efficiency gains across the three reform stages

The three sequential reforms (ISP15 settlement, MTU15-IDA clearing, MTU15-DA clearing) each delivered a distinct efficiency channel. The next table documents them side by side, with the outcome variable explicitly specified for each (per the never-abstract-effect rule):

Table 35: Efficiency gains per reform stage, by outcome. Reform dates: ISP15 = 2024-12-01 (settlement granularity 60-min $\rightarrow$ 15-min); MTU15-IDA = 2025-03-19 (IDA clearing 60-min $\rightarrow$ 15-min); MTU15-DA = 2025-10-01 (DA clearing 60-min $\rightarrow$ 15-min, overlaid with operational-reforzada). Per-day costs from Table 34; imbalance metrics from ESIOs indicators 762, 1338, 686, 687.

Reform	Outcome variable	Measured change
ISP15 (Dec 2024)	Mean abs imbalance vol / cell	~ 810 MW (hourly, 3-sess) $\rightarrow \sim 325$ MW (15-min, ISP15-win)
	Net of variance-averaging	$\sim 50\%$ behavioural reduction
	Dual-price spread (cobro-up / pago-dn)	40/85 $\rightarrow$ 70/110 EUR/MWh (wider, more punitive)
	DA spot price	no causal change (settlement reform, not clearing)
MTU15-IDA (Mar 2025)	TR (tiempo real) up price	€545/MWh $\rightarrow$ €226 (-59%)
	TR-up cost per day	€5.8M $\rightarrow$ €2.9M (-50%)
	DA spot price level	no clean causal change (overlap w/ reforzada)
	IDA clearing within hour	4 separate prices per hour (mechanical)
MTU15-DA (Oct 2025)	TR-up cost per day (further drop)	€2.9M $\rightarrow$ €1.8M (-38%)
	aFRR-capacity (banda) price	€13/MWh $\rightarrow$ €6/MWh (-54%)
	Fase I-up cost per day	€10.5M $\rightarrow$ €13.0M ($+24\%$ reforzada-driven)
	Total ajuste cost per day	€17.7M $\rightarrow$ €21.1M ($+19\%$ net)

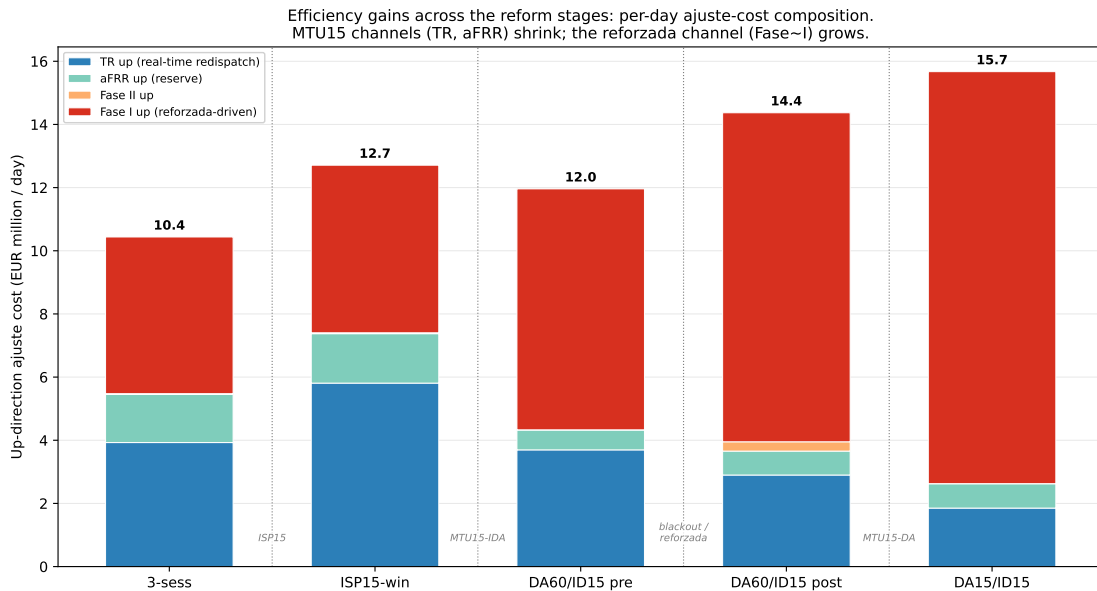


Figure 17: Per-day up-direction ajuste-cost composition across the five reform-window regimes. Stacked bars: per-day cost (EUR million) of the four up-direction channels — TR (real-time redispatch), aFRR reserve, Fase II, Fase I. Bar height = total up-direction ajuste cost per day; the number above each bar is that total. The MTU15 efficiency channels (TR up, aFRR up) shrink across the reform sequence — TR up from ~ 5.8 M/day at ISP15-win to ~ 1.9 M/day at DA15/ID15 — while the reforzada channel (Fase I up) grows from ~ 5 M/day to ~ 13 M/day. The net per-day cost rises because the reforzada-driven Fase I growth outweighs the granularity-driven TR/aFRR savings; the two are logically distinct policy effects (§12) that happen to overlap in calendar time. Window-length-normalised from `per_regime_costs_official.csv`.

Reading the efficiency gains.

- **ISP15 is the cleanest reform identification.** Outcome = absolute imbalance volume per ISP cell; the 60min $\rightarrow$ 15min settlement granularity mechanically shrinks per-cell variance by $\sim 4\times$, so $\sim 50\%$ of the observed $\sim 60\%$ reduction is measurement and the residual ~ 10 pp is behavioural. The dual-price spread also widens (cobro/pago move

apart by 30/25 EUR/MWh, more punitive). No overlapping reform in the Dec 2024–Mar 2025 window, so this is the cleanest single-shock identification.

- **MTU15-IDA delivered the big TR-cost channel.** TR cost dropped from €5.8M to €2.9M per day — the largest single efficiency gain in the full cascade. Mechanism: granular IDA absorbs intra-hour scarcity that previously bled into real-time post-clearing intervention. TR price halved precisely because the residual demand for real-time correction shrank.
- **MTU15-DA delivered a smaller marginal TR + aFRR efficiency channel, but was contemporaneous with reforzada-driven Fase I cost growth.** The marginal TR savings (an additional €1.1M/day) and the aFRR-capacity price collapse are clean MTU15-DA effects: nothing in the operational-reforzada policy mechanism would drive these. The Fase I cost growth (€2.5M/day extra) is a SEPARATE policy effect from reforzada (tighter PO-3.x security criteria force more CCGTs on, paid pay-as-bid). The two are confounded in the post-Oct 2025 window but logically distinct. The net is +19% per-day cost, but the decomposition is:
 - Genuine MTU15-DA efficiency channel: $-\text{€}1.1\text{M/day (TR)} + \text{smaller aFRR savings} \approx -\text{€}1.5\text{M/day}$
 - Reforzada cost growth (independent of MTU15-DA): $+\text{€}2.5\text{M/day (Fase I)}$
 - Other components (Fase II, RES curtailment): small

The MTU15-DA reform did deliver an efficiency dividend, but it is swamped in the aggregate cost data by the parallel reforzada cost growth. Separately identifying the two requires either the 40-day DA60/ID15 pre-blk window (post-MTU15-IDA, pre-reforzada) or a geographic/firm-level discontinuity (non-reforzada zones, non-required CCGTs).

15 Cournot reading and CNMC enforcement context

15.1 Cournot reading of the cap-padded ladder

A near-constant DA bid price ladder at the cap (GN at ~ 840 EUR/MWh in Table 31) combined with all the adjustment on the *quantity* side — the unit either clears DA or it does not, but its bid price hardly varies — is the empirical signature of a **Cournot strategy on the quantity margin**. In pure economic terms this is equivalent to direct price-raising in DA, but the regulatorily defensible posture for the firm is “my offer price has not changed; my bids follow my published ladder.” The redispatch channel (Fase I pay-as-bid) is then where the per-MWh rent is captured.

15.2 Historical CNMC sanction precedent

The empirical pattern documented in §§11–13 has a direct precedent in CNMC enforcement. The resolutions for the relevant cases are archived in the project’s regulation document folder.

Table 36: CNMC sanction resolutions relevant to the practice. “LSE” = Ley del Sector Eléctrico 24/2013. Sources: CNMC resolution PDFs archived in the project regulation folder.

Year	Firm / Plant	Expediente	LSE article	Sanction
2015	Iberdrola hydro	SNC-DE-046-14	64 muy grave (manipulation)	25.0 M EUR
2019	Naturgy CCGT [†]	SNC-DE-175-17	65.34 grave (abnormal DA offers)	~7 M EUR
2019	Endesa Besós 3	SNC-DE-174-17	65.34 grave (abnormal DA offers)	6.6 M EUR
2023	Naturgy Sabón 3 (Galicia)	—	65.33 grave (RT abuse)	confidential
2023	Engie Castelnou	SNC-DE-152-22	grave (availability)	—
2023	Ignis Escatrón 2	SNC-DE-151-22	grave (availability)	—

[†] Plants in scope: Besós 4, PBCN 1/2, Sagunto 1/2/3, Málaga 1, SROQ 1 — exactly the Cataluña / Levante / Sur zonal map (§13). CNMC-estimated minimum benefit 13.0 M EUR over 4 months.

Reading. The 2019 Naturgy sanction (SNC-DE-175-17) is the smoking-gun precedent: same plants, same zones, same mechanism (high DA → unit not despatched in DA → unit systematically programmed at higher RT prices) over October 2016 – January 2017. The 2024–26 evidence in this Part B is the same practice continuing post-*operación reforzada*, now at fleet-wide scale, with the within-hour granularity reform as an additional tool. The 2023 Sabón 3 sanction extends the geography to Galicia (the GE / GN duopoly per §13).

15.3 Blackout-cohort expedientes (separate channel)

In April 2026 CNMC opened a parallel set of ~ 56 expedientes (IB 22, GE 17, GN 5, others) under Art 65.8 of the LSE, for **voltage-control violations** on April 28, 2025 (the day of the blackout). This is a *distinct regulatory channel*: it concerns failure to provide reactive support during the blackout sequence, not auction-shaping in DA. We flag it for completeness; the two channels are not conflated here.